DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS

Question Bank with Model Answers

Course Title	Entrepreneurship and Intellectual Property Rights	**Course Code**	HS351TA
Semester	V	**Units Covered**	Unit I and Unit IV (patents half)
Compiled From	CIE and SEE papers, 2018–2026 and the faculty question bank	**Questions**	Part A: 70 Part B: 59

Notes:

1. Questions are compiled from the CIE and SEE question papers of 2018–2026 and the faculty question bank. The unit and the source paper are shown below each question.
2. Part A contains fill-in-the-blank and short-answer questions. Part B contains descriptive, case-based and application questions (5 to 10 marks).
3. Model answers are written to the depth expected for the marks allotted. Numerical answers show the complete working, and the final result is stated after **Answer:**.
4. Where a question repeats an earlier one, the answer refers to the earlier question (the reference is a link).
5. M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome. Only questions answerable from the Unit I notes and from the patents half of the Unit IV notes (introduction to IP, patents, traditional knowledge and biodiversity) are included, taken from the CIE and SEE papers and from the faculty question bank; a dash means the paper did not print that value.

Sl. No.	Questions	M	BT	CO
	PART – A			
1	Define: Entrepreneur. Source: Unit I | SEE Feb/Mar 2025, Q1.2; Question Bank, Unit 1 short answers, Topic 4 Q3	02	2	1

An **entrepreneur** is an individual who identifies an opportunity, assumes the risks and responsibilities of forming a new business, and organises resources (people, money, technology) to create value and earn profit. The word comes from the French entreprendre, meaning to undertake. An entrepreneur is an innovator, creator and risk-taker who introduces something new to the economy: a new product, process, source of material or market.

2	The term entrepreneur is derived from the term "entreprendre" which means _____. (The word entrepreneur has been taken from the _____ language.) Source: Unit I | SEE Apr/May 2024 (21HS51A), Q1.5; SEE Nov/Dec 2019 (16HSI51), Q1.15	01	1	4

Answer: French; entreprendre means to undertake

An entrepreneur is one who undertakes or plans for all the risks and responsibilities that come with the formation of a new business in order to earn profit.

3	List any two characteristics of successful entrepreneurs. Source: Unit I | CIE-I, 25/11/2024; CIE-I, 07/11/2025; SEE Mar/Apr 2023 (18HSI51), Q1.5; SEE Mar/Apr 2023 (16HSI51), Q1.6; Question Bank, Unit 1 short answers, Topic 3 Q3; Topic 4 Q7	02	2	1

- **Self-motivation:** pushes on and stays dedicated to the plan even without an immediate pay cheque or anyone to supervise.
- **Calculated risk-taking:** accepts risks that are more likely to pay off, because playing safe seldom leads to success.

Other characteristics: vision, leadership, flexibility, creativity, passion, networking ability and basic money-management skills.

4	List four types of entrepreneurs. Source: Unit I | SEE Feb/Mar 2025, Q1.1; CIE-I, 07/11/2025	02	1	1

(1) **The inventor:** creates new products or business ideas that do not yet exist.
(2) **The small business owner:** runs enterprises with fewer than 500 workers, from consultancies to local restaurants.
(3) **The online entrepreneur:** sells products, runs blogs or develops software online and deals directly with clients through the internet.
(4) **The home business owner:** uses the home as the base of the business, for example a soap maker working from a basement.

Note: Entrepreneurs are also classified as innovative, imitative, Fabian and drone entrepreneurs (see Part B Q4).

5	Rahul, a software engineer, notices that many small businesses struggle with online payments. He develops a mobile app to simplify digital transactions and starts his own company to offer this solution. Identify the type of entrepreneur Rahul represents and explain why. Source: Unit I | CIE-I, 07/11/2025	02	3	1

Answer: Rahul is an innovative (online, technology-based) entrepreneur.

He identified a gap in the market (the difficulty small businesses face with online payments), created a new product (a mobile app that simplifies digital transactions) and launched his own company around it. Introducing a new product or service to meet an unmet need is the mark of an innovative entrepreneur, and because the offer is software delivered online he also fits the online entrepreneur and innovation entrepreneurship models.

6	Differentiate between an entrepreneur and an intrapreneur. Source: Unit I | CIE-I, 25/11/2024; SEE Feb/Mar 2026, Q1.2; Question Bank, Unit 1 short answers, Topic 4 Q1, Q9	02	1	1

Aspect	Entrepreneur	Intrapreneur
Definition	An individual who starts a new venture.	An innovator working within an existing organisation.
Risk	Takes personal financial and business risk.	Risk is borne by the organisation.
Resources	Builds resources independently.	Uses organisational resources.
Autonomy	Operates independently.	Functions under organisational policies.
Reward	Retains all profits or losses.	Shares rewards with the company.

7	List any two myths about entrepreneurship. Source: Unit I | CIE-I, 25/11/2024; SEE Jun 2023 Makeup (18HSI51), Q1.14; SEE Mar/Apr 2023 (16HSI51), Q1.9; SEE Jan/Feb 2021 (16HSI51), Q1.14; SEE Jul 2019 Fast Track (16HSI51), Q1.13; SEE Nov/Dec 2018 (16HSI51), Q1.6; Question Bank, Unit 1 short answers, Topic 4 Q4	02	2	1

- **Entrepreneurs are born that way:** in reality entrepreneurship is a learned skill; almost anyone can acquire the necessary knowledge and skills.
- **The only requirement is a good idea:** even the best idea needs execution, planning, talent, leadership and communication.

Other myths: starting a business guarantees freedom; it quickly leads to wealth; only money motivates employees; businesses either flourish or fail; all responsibility falls on the entrepreneur; there is a silver bullet to success; an MBA is needed at the helm; quitting is for losers.

8	The term _____ is used for corporate entrepreneurship. Source: Unit I | SEE Jul 2019 Fast Track (16HSI51), Q1.11	01	–	–

Answer: Intrapreneurship (intrapreneur)

Corporate entrepreneurship encourages employees of an established firm to think and act like entrepreneurs and to innovate inside the organisation, using the firm's resources and bearing little personal risk.

9	_____ is the means by which the entrepreneur either creates new wealth-producing resources or endows existing resources with enhanced potential for creating wealth. Source: Unit I | SEE Jul 2019 Fast Track (16HSI51), Q1.12	01	–	–

Answer: Innovation

Innovation is the ability to create new wealth-producing resources or to enhance the potential of existing ones: new product lines, better quality, new ideas, technologies and markets.

10	Jack, a college final-year student, is from a well-off business family. He has spotted an opportunity in his hometown related to the production of electricity using wind. Can he become an entrepreneur and turn this into a venture? Source: Unit I | SEE Jan/Feb 2021 (18HSI51), Q1.12; SEE Mar/Apr 2023 (16HSI51), Q1.10	01	–	–

Answer: Yes. Jack can become an entrepreneur.

Entrepreneurship is a learned skill, not a natural-born ability, and it does not need an MBA or a particular background. Jack has already done the first thing an entrepreneur does, which is to spot an opportunity (wind power in his hometown). His business family gives him exposure and possible support, but to turn the idea into a venture he must plan, arrange resources, build a team and take calculated risks.

11	Why are small businesses important to a country's economy? Source: Unit I | SEE Mar/Apr 2023 (18HSI51), Q1.6	02	–	–

- **Job creation:** they generate local employment and income for families.
- **Source of large firms:** all big industrial organisations usually begin as small businesses; about 75 per cent of businesses fall in the small-enterprise model.
- **Meeting specific needs and community development:** they serve niche local demand, keep competition alive and add to national income and the well-being of the community.

12	How does entrepreneurship contribute to economic growth and national development? Source: Unit I | Question Bank, Unit 1 short answers, Topic 1 Q1; Topic 3 Q8	02	1	1

Entrepreneurship raises national income and GDP, creates jobs, and drives innovation and industrial development. Entrepreneurs invest their own and mobilised savings, open new markets, pay taxes, and spread business opportunity across regions, which supports national development.

13	State any two scopes of entrepreneurship. Source: Unit I | Question Bank, Unit 1 short answers, Topic 1 Q2	02	2	1

- **Creation of employment opportunities:** new ventures create jobs at every level.
- **Development of new products and markets:** innovation in goods, services and processes opens new demand.

The scope also spans commercial, social, agricultural, technological and corporate ventures.

14	Define entrepreneurship. Source: Unit I | Question Bank, Unit 1 short answers, Topic 1 Q3	01	2	1

Entrepreneurship is the process of designing, launching and running a new business, typically starting as a small enterprise offering a product, process or service. It involves identifying an unmet need, creating a solution and organising resources to deliver value, with opportunity recognition and risk-taking.

15	Explain how entrepreneurship contributes to economic growth and engineering innovation. Source: Unit I | Question Bank, Unit 1 short answers, Topic 1 Q4	02	3	1

Entrepreneurship drives economic growth by creating employment, raising productivity and introducing technological innovations. In engineering it encourages innovation through product development, process improvement and the commercialisation of new technologies, so that engineers turn inventions into enterprises.

16	Identify one analytical method used for opportunity identification. Source: Unit I | SEE Feb/Mar 2026, Q1.1; Question Bank, Unit 1 short answers, Topic 2 Q1, Q4	02	1	1

Answer: SWOT analysis (Strengths, Weaknesses, Opportunities, Threats)

It compares the internal strengths and weaknesses of the entrepreneur with the opportunities and threats in the market, so that ideas with a good fit can be picked.

17	Give one example of an innovative entrepreneur and one of an imitative entrepreneur. Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q2	02	3	1

- **Innovative entrepreneur:** Steve Jobs (Apple), who introduced new products such as user-friendly personal computers and the iPhone.
- **Imitative entrepreneur:** one who adopts a proven model with local changes, for example an online marketplace set up on the lines of the Amazon model.

18	What is meant by an entrepreneurial opportunity? Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q3	02	3	1

An **entrepreneurial opportunity** is a favourable situation, usually an unmet need or market gap, that allows an entrepreneur to introduce a new product, service or process to the market for profit.

19	What does the abbreviation SWOT stand for? Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q5	02	3	1

Answer: Strengths, Weaknesses, Opportunities and Threats

20	Which technique involves generating many creative ideas without immediate evaluation? Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q6	02	3	1

Answer: Brainstorming

A group generates many creative ideas without judging them at once; evaluation follows later.

21	What is environmental scanning in entrepreneurship? Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q7	02	3	1

Environmental scanning is the process of analysing external factors such as market trends, technology, competition, regulation and social and economic changes in order to find potential opportunities and threats for a business.

22	Name a technique that focuses on customer problems to identify opportunities. Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q8	02	3	1

Answer: Problem identification (customer need analysis)

The entrepreneur observes the problems customers face and designs a product or service to solve them.

23	What is the purpose of market research in opportunity identification? Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q9	02	3	1

Market research gathers information on customer needs and preferences, gaps in the market and competitor offerings, which shows whether an idea has real demand before resources are committed.

24	Which analytical tool helps in comparing business ideas based on feasibility and potential return? Source: Unit I | Question Bank, Unit 1 short answers, Topic 2 Q10	02	3	1

Answer: Feasibility analysis (an opportunity evaluation matrix)

It scores each idea on market demand, technical and financial feasibility and expected return so that the best opportunity can be chosen.

25	Analyse how networking skills influence entrepreneurial success. Source: Unit I | Question Bank, Unit 1 short answers, Topic 3 Q1	02	2	1

Strong networks give access to investors, mentors, suppliers, customers and partners. They help in raising funds, recognising partnership opportunities, finding talent and getting market information, and a good network can take an entrepreneur a long way.

26	How do sustainable start-ups contribute to modern entrepreneurship? Source: Unit I | Question Bank, Unit 1 short answers, Topic 3 Q2	02	2	1

Sustainable start-ups build environmental and social goals (ESG principles) into the business: eco-friendly products, green supply chains, renewable energy and solutions to problems such as climate change and poverty. They open new markets, attract conscious customers and investors, and make growth responsible.

27	How does self-confidence contribute to entrepreneurial success? Source: Unit I | Question Bank, Unit 1 short answers, Topic 3 Q4	02	2	1

Self-confidence lets the entrepreneur trust his or her own decisions, overcome challenges and setbacks, and inspire investors, employees and customers to believe in the vision.

28	What is green entrepreneurship? Source: Unit I | Question Bank, Unit 1 short answers, Topic 3 Q5	02	2	1

Green entrepreneurship is the creation of environmentally sustainable products and services that reduce ecological impact, for example solar products, recycling ventures and eco-friendly packaging.

29	Mention any two emerging trends in modern entrepreneurship. Source: Unit I | Question Bank, Unit 1 short answers, Topic 3 Q6	02	2	1

- **Technology-driven and digital entrepreneurship:** AI, data analytics, e-commerce and fintech.
- **Social and sustainable entrepreneurship:** ventures that put environmental and social impact beside profit.

Others: remote work and virtual teams, agile business models, globalisation, and growing participation of women, youth and rural entrepreneurs.

30	Differentiate between an entrepreneur and entrepreneurship. Source: Unit I | Question Bank, Unit 1 short answers, Topic 3 Q7	02	2	1

Basis	Entrepreneur	Entrepreneurship
Meaning	The individual who starts and manages a business.	The process or activity of starting and managing a business venture.
Nature	A person.	A process.

31	Analyse one key reason for the rise of Indian start-ups. Source: Unit I | Question Bank, Unit 1 short answers, Topic 4 Q2	01	2	1

The availability of **venture capital** and a supportive policy ecosystem (start-up schemes, incubators and easier regulation) encourage innovation and let new ideas get funded and grow. A large market and digital infrastructure also help.

32	Who is an intrapreneur? Source: Unit I | Question Bank, Unit 1 short answers, Topic 4 Q5	01	2	1

An **intrapreneur** is an employee within an organisation who innovates and develops new ideas as if running his or her own business, using the organisation's resources and with the risk borne by the organisation.

33	What is the main role of an entrepreneurial team? Explain its significance in new venture creation. Source: Unit I | Question Bank, Unit 1 short answers, Topic 4 Q6, Q10	02	2	1

The team combines diverse skills and perspectives, shares responsibilities and works collaboratively to achieve business success. In new-venture creation it gives complementary skills, better decisions through shared problem-solving, a shared workload, and greater innovation, sustainability and investor confidence.

34	How do myths about entrepreneurship affect aspiring entrepreneurs? Source: Unit I | Question Bank, Unit 1 short answers, Topic 4 Q8	02	2	1

Myths create false beliefs that discourage people from starting ventures, for example that one needs huge capital, an MBA or a natural gift. They can also lead to wrong decisions: expecting quick wealth, trying to do everything alone, or not quitting an idea that does not work, which causes disappointment and burnout.

35	What is a patent? Source: Unit IV | SEE Jan/Feb 2021 (18HSI51), Q1.1; Question Bank, Unit 4 Part A Q2	02	–	–

A **patent** is an exclusive right granted for a new invention (a machine, process, article of manufacture or composition of matter) that is new, involves an inventive step and is capable of industrial application. The patentee can prevent others from making, using, selling, offering for sale or importing the invention without permission, generally for 20 years from the date of filing the application.

36	List the criteria of patentability. (An invention has to be _____ and _____ in order to be patentable.) Source: Unit IV | SEE Jan/Feb 2021 (18HSI51), Q1.2; SEE Apr 2024 (18HSI51), Q1.3; SEE Nov/Dec 2019 (16HSI51), Q1.5	02	2	2

(1) **Novelty:** the invention must be new, i.e. not part of the prior art (not published or used anywhere before the date of filing).
(2) **Inventive step:** it must involve a technical advance or economic significance and be non-obvious to a person skilled in the art.
(3) **Industrial application:** it must be capable of being made or used in an industry (utility).

Answer: An invention has to be new (novel) and useful (capable of industrial application, with an inventive step).

37	Define compulsory licensing under the patent law. Source: Unit IV | CIE-I, 07/11/2025; CIE-I, 25/11/2024; Question Bank, Unit 4 Part A Q18	02	2	4

A **compulsory licence** is a licence granted by the Controller of Patents to a third party to work a patented invention without the consent of the patentee. An application can be made at any time after three years from the sealing of the patent on the grounds that (i) the reasonable requirements of the public are not satisfied, (ii) the invention is not available at a reasonably affordable price, or (iii) it is not worked in India. It can also be granted on notification by the Central Government in a national emergency, extreme urgency or public non-commercial use.

38	A compulsory licence will be granted only after the expiry of _____ years from the date of sealing of the patent. Source: Unit IV | SEE Nov/Dec 2018 (16HSI51), Q1.1; SEE Nov/Dec 2019 (16HSI51), Q1.13	01	–	–

Answer: Three (3) years

39	In India, _____ is issued if the patented article is not available to the public at an affordable price. Source: Unit IV | SEE Jul 2019 Fast Track (16HSI51), Q1.4	01	–	–

Answer: A compulsory licence

The unaffordable price of the patented article is a ground on which the Controller can order the patentee to grant a licence to the applicant.

40	List any two non-patentable inventions under the Indian Patents Act. Source: Unit IV | CIE-I, 25/11/2024; Question Bank, Unit 4 Part A Q7	02	2	2

- Inventions relating to atomic energy.
- A mathematical or business method, a computer programme per se, or an algorithm.

Others: frivolous inventions or those contrary to natural laws (perpetual motion machine), mere discovery of a scientific principle, methods of agriculture or horticulture, and processes for medical treatment of humans and animals.

41	Are inventions relating to atomic energy patentable? Source: Unit IV | SEE Apr 2024 (18HSI51), Q1.9	01	1	2

Answer: No. Inventions relating to atomic energy are not patentable.

The Central Government has the sole responsibility for the development of atomic energy and does not want its programmes hampered by patent disclosure and claims.

42	The term of an Indian patent is _____ years from the date of filing of the application. Source: Unit IV | SEE Apr 2024 (18HSI51), Q1.11; SEE Jan/Feb 2021 (16HSI51), Q1.4	01	1	2

Answer: Twenty (20) years

The term runs from the date of filing the application. The patent must be renewed every year on payment of a fee; failure to renew results in loss of all patent rights.

43	If a provisional specification is filed, the complete specification must be filed within _____ months. Source: Unit IV | SEE Jun 2023 Makeup (18HSI51), Q1.6; SEE Jan/Feb 2021 (16HSI51), Q1.3	01	–	–

Answer: Twelve (12) months

The complete specification must be filed within one year of filing the application with the provisional specification; otherwise the application is deemed to be abandoned.

44	Mention any two objectives of the patent system. Source: Unit IV | SEE Feb/Mar 2026, Q1.7; Question Bank, Unit 4 Part A Q5	02	2	4

- To encourage inventions and to secure that they are worked in India on a commercial scale without undue delay, and not merely to give the patentee a monopoly of importation.
- To promote technological innovation and the transfer and dissemination of technology for social and economic welfare, while protecting public health and making the invention available at reasonably affordable prices.

45	When the patentee agrees to give another person certain defined rights in the patent with immediate effect by a document and not by an agreement, the assignment is termed as _____. Source: Unit IV | SEE Apr/May 2024 (21HS51A), Q1.1	01	2	1

Answer: Equitable assignment

It cannot be registered in the Register of Patents until it is converted into a legal assignment by a written, registered document.

46	An assignment of an existing patent through an _____ which has been duly registered is called a legal assignment. Source: Unit IV | SEE Apr/May 2024 (21HS51A), Q1.2	01	1	2

Answer: Agreement

The assignee's name is then entered in the Register of Patents as the proprietor.

47	Can the assignee of a true inventor apply for a patent? Source: Unit IV | SEE Apr/May 2024 (21HS51A), Q1.3	01	4	1

Answer: Yes. The assignee of the true and first inventor can apply for the patent.

An application may be made by the true and first inventor, by his assignee, or by the legal representative of a deceased inventor. The assignee must file proof of the right to apply, and the inventor must still be named.

48	Through which mode can the Central Government acquire a patent from the patentee for a public purpose? Source: Unit IV | SEE Jan/Feb 2021 (16HSI51), Q1.9	01	–	–

Answer: Operation of law

The Central Government can acquire a patent or application by a notification in the Official Gazette when it is necessary for a public purpose; all rights then stand transferred to it, with compensation mutually agreed to the patentee.

49	If somebody wants to develop a new method of horticulture, can he get a patent for it? (Is a method of growing plants on a rooftop patentable?) Source: Unit IV | SEE Nov/Dec 2019 (16HSI51), Q1.3; SEE Jun 2023 Makeup (18HSI51), Q1.4	02	–	–

Answer: No. A method of agriculture or horticulture is not patentable.

Under the Indian Patents Act these methods are not treated as inventions, as a matter of policy. A method of growing plants on a rooftop falls in the same category.

50	If a person invents a new process for recording music, he or she will likely apply for a _____. Source: Unit IV | SEE Nov/Dec 2019 (16HSI51), Q1.1; SEE Apr 2024 (18HSI51), Q1.2	01	2	1

Answer: Patent

A new process (a method of recording music) is a technical invention that can be patented if it is new, inventive and industrially applicable. The music itself is a creative work protected by copyright.

51	A composer, Sajan, composed music for a Kannada film. Can he apply for a patent for the music he composed? Why? Source: Unit IV | SEE Nov/Dec 2018 (16HSI51), Q1.8	02	–	–

Answer: No. Music is protected by copyright, not by a patent.

A musical work is an artistic creation. Patents are available only for inventions, that is, new products or processes that involve an inventive step and are capable of industrial application; musical and other artistic works are excluded from patents and covered by copyright.

52	Software is protected as intellectual property in India primarily under the law of _____. Source: Unit IV | SEE Apr 2024 (18HSI51), Q1.10	01	2	1

Answer: Copyright (the Copyright Act, 1957)

A computer programme per se is not patentable in India.

53	How do you protect computer software that has a functional relationship with the computer in India? Source: Unit IV | SEE Jul 2019 Fast Track (16HSI51), Q1.5	01	–	–

A computer programme per se is protected under the Copyright Act, not by a patent. However, when a programme enables the computer to operate in a new way to produce a required result, the computer so programmed is a modified equipment and the system (the computer along with the programme) is **patentable**.

54	A person made minor improvements to a patent-expired drug used to treat leukaemia or cancer, making it easier to swallow. Is he eligible to hold a fresh patent on the drug with the minor improvement? Source: Unit IV | SEE Jul 2019 Fast Track (16HSI51), Q1.1	02	–	–

Answer: No, the minor improvement does not qualify for a fresh patent.

Once the patent has expired the drug falls into the public domain. Making a known drug easier to swallow is only a new form of a known substance that does not enhance its known efficacy; the Act treats such new forms, and derivatives that do not differ significantly in efficacy, as the same substance and denies patent protection. A fresh patent would need a genuinely new, inventive and useful result.

55	Can an employer apply for a patent over the intellectual effort of his employee? Source: Unit IV | SEE Jul 2019 Fast Track (16HSI51), Q1.6	01	–	–

Answer: Yes, but only as the employee's assignee under the employment contract.

The invention made during employment is patentable in the name of the employee as inventor. The ownership of the patent depends on the contract between employee and employer, and in R&D organisations the inventions are commonly assigned to the employer while the contributors are still listed as inventors.

56	A scientist gets an idea on how to make biodegradable plastics. How can this idea be patented by the scientist? Source: Unit IV | SEE Mar/Apr 2023 (16HSI51), Q1.1	02	–	–

An idea alone cannot be patented. The scientist must reduce it to a definite conception of the complete and operative invention (a new biodegradable plastic or its method of production) that is new, involves an inventive step and is capable of industrial application. He should then file one application in the prescribed form and fee with a provisional specification, which fixes the priority date, and file the complete specification with claims within twelve months. The application then goes through publication, examination and grant.

57	If a company develops a new technology that improves its main product, what type of intellectual property can it use to stop others from copying the invention? Source: Unit IV | SEE Jun 2023 Makeup (18HSI51), Q1.5	01	–	–

Answer: Patent

A patent gives the owner the exclusive right for 20 years to prevent others from making, using, selling or importing the new technology without permission.

58	Traditional knowledge is limited to any specific field of knowledge. State True or False. Source: Unit IV | SEE Jun 2023 Makeup (18HSI51), Q1.2	01	–	–

Answer: False

Traditional knowledge covers a wide range of fields: knowledge of plant genetic resources and the adaptation of crops to soils and climate, medicinal uses of plants and animals, life technologies and cultural products, passed on from generation to generation usually by word of mouth within a group.

59	What is Intellectual Property (IP)? Source: Unit IV | Question Bank, Unit 4 Part A Q1	02	2	4

Intellectual property refers to creations of the mind such as inventions, literary and artistic works, designs, symbols, names and images used in commerce. The creator or owner is granted exclusive rights, called intellectual property rights, so that he benefits financially and the work is protected.

60	What are patentable inventions? Source: Unit IV | Question Bank, Unit 4 Part A Q6	02	2	4

Patentable inventions are new products or processes that are novel, involve an inventive step (are non-obvious to a person skilled in the art) and are capable of industrial application, for example machines, processes and chemical compositions.

61	Rahul invents a new type of solar-powered mobile charger that charges devices twice as fast as conventional chargers. Is Rahul's invention patentable? Why? Source: Unit IV | Question Bank, Unit 4 Part A Q8	02	3	4

Answer: Yes, Rahul’s invention is patentable.

A solar charger that charges twice as fast as conventional chargers is a new technical product with a technical advance (an inventive step) and is capable of industrial application, and it does not fall under any excluded category. It must be new in the world, that is, not already published or sold.

62	Dr. Meena discovers a new mathematical formula for predicting weather patterns. Can this discovery be patented? Source: Unit IV | Question Bank, Unit 4 Part A Q9	02	3	4

Answer: No, a mathematical formula cannot be patented.

A mathematical method is an abstract idea and a mere discovery of a scientific principle, which the Indian Patents Act does not treat as an invention. It would qualify only if embodied in a technical process with an industrial result.

63	A pharmaceutical company develops a new process to manufacture an existing drug more efficiently. Is this invention patentable? Source: Unit IV | Question Bank, Unit 4 Part A Q10	02	3	4

Answer: Yes, a new process is patentable.

The drug itself is known, but a new manufacturing process that is novel, inventive and industrially applicable can be granted a process patent. The patentee can then prevent others from using the process and from selling the product directly obtained by it.

64	Ravi finds a naturally occurring medicinal plant that cures skin allergies and wants to patent it. Can Ravi get a patent for this? Source: Unit IV | Question Bank, Unit 4 Part A Q11	02	3	4

Answer: No, Ravi cannot get a patent.

Finding a naturally occurring plant is a mere discovery of a living thing occurring in nature, not an invention. A new use of the plant is also not patentable unless a new, non-obvious technical product or process is developed.

65	A researcher designs a new variety of rice through cross-breeding. Is this patentable under the Patent Act? Source: Unit IV | Question Bank, Unit 4 Part A Q12	02	3	4

Answer: No, a new plant variety is not patentable.

Plants and animals in whole or part (other than micro-organisms), including seeds, varieties and species, and essentially biological processes such as cross-breeding, are excluded. A new variety can be protected by the separate (sui generis) system for plant varieties and breeders' rights.

66	An inventor develops a robot that automatically cleans solar panels to improve efficiency. Is this invention patentable? Source: Unit IV | Question Bank, Unit 4 Part A Q13	02	3	4

Answer: Yes, the robot is patentable.

It is a new, useful machine that solves a technical problem (cleaning solar panels automatically to improve efficiency) and is capable of industrial application, so it meets novelty, inventive step and industrial applicability.

67	A researcher files a patent for turmeric's healing properties, which are already known in India's traditional medicine. Why was this patent rejected? Source: Unit IV | Question Bank, Unit 4 Part A Q13 (second)	02	3	5

The patent was rejected because the healing property of turmeric was already known in India's traditional medicine for centuries. It was therefore part of the prior art and public knowledge and lacked novelty, and traditional knowledge cannot be patented. (In the actual case the US patent was revoked after CSIR produced published evidence.)

68	A foreign firm files for a patent in India for neem-based pesticide technology already in public use. What will the Indian Patent Office decide? Source: Unit IV | Question Bank, Unit 4 Part A Q15	02	3	5

The Indian Patent Office will reject it. Neem's pesticidal use is traditional knowledge already in public use, so the technology lacks novelty, and an invention which in effect is traditional knowledge is not patentable.

69	The Indian government allows a local company to produce a patented medicine during a health emergency without the owner's consent. What is this called under patent law? Source: Unit IV | Question Bank, Unit 4 Part A Q16	02	3	5

Answer: Compulsory licensing

The Government (through the Controller, or on notification in a national emergency or extreme urgency) authorises a third party to produce or sell the patented medicine without the patentee's consent.

70	A government agency documents and publishes local medicinal practices to prevent foreign patent claims. What is the purpose of this initiative? Source: Unit IV | Question Bank, Unit 4 Part A Q17	02	4	5

The purpose is to **protect traditional knowledge and prevent biopiracy**. Published documentation creates prior art that patent examiners can search, so foreign firms cannot obtain patents on knowledge already known, and the rights of local communities are safeguarded.

Sl. No.	Questions	M	BT	CO
	PART – B			
1	Justify the importance of entrepreneurship. Source: Unit I | SEE Feb/Mar 2025, Q2a	08	5	1

Entrepreneurship is the art of starting a business that offers a creative product, process or service. The entrepreneur, from the French entreprendre (to undertake), takes the risks and responsibilities of forming a new business to earn profit. Its importance is seen in the following benefits.

(1) **Creation of job opportunities:** new firms create jobs at every level; the entrepreneur earns for himself and employs others.
(2) **Creation of new businesses:** every large organisation began as a small initiative, and entrepreneurship keeps new ventures coming.
(3) **Innovation:** it generates new wealth-producing resources or improves existing ones, creating new product lines, better quality, technologies and markets.
(4) **Better standard of living:** a wide range of goods and services is made available, and the income generated for families raises living standards.
(5) **Support to research and development:** entrepreneurs fund and work with research institutions, which helps discoveries.
(6) **Community development:** goodwill, charitable support and better infrastructure, sanitation and a skilled workforce improve quality of life.
(7) **Increased productivity:** new entrants force existing firms to cut costs, improve quality and offer more value for money.
(8) **Creation of national wealth:** entrepreneurs invest, open new markets and pay taxes, adding to the gross domestic product and national income.
(9) **Social welfare:** income for poor and marginalised sections helps them meet basic needs and creates social integration.

Conclusion: entrepreneurship converts ideas into enterprises, and through jobs, innovation and wealth creation it drives both individual and national progress.

2	Explain the importance of entrepreneurship in promoting engineering innovation and fostering economic growth. Discuss how entrepreneurial initiatives among engineers contribute to technological advancement, job creation and sustainable development, with suitable examples. Source: Unit I | CIE-I, 07/11/2025	10	3	1

Entrepreneurs turn technical knowledge into products and enterprises. For engineers, entrepreneurship is the route by which innovations leave the laboratory and reach the market, and in doing so it grows the economy.

1. Engineering innovation

- Entrepreneurs introduce new technologies, products and services and open new markets; engineers who become entrepreneurs bring the technical depth to do it.
- Alexander Graham Bell, with Thomas Watson, built the first voice-transmitting device, the telephone, and founded the Bell Telephone Company. Jeff Bezos, who studied electrical engineering and computer science, built Amazon; Larry Page and Sergey Brin built the Google search engine. Apple and Tesla are examples of innovation entrepreneurship.

2. Technological advancement

- Large entrepreneurial firms such as Apple and Samsung invest constantly in R&D to advance their technology.
- Today's ventures use artificial intelligence, data analytics, IoT, blockchain and digital platforms; e-commerce, fintech and software-as-a-service businesses are redefining traditional industries.

3. Job creation

- Start-ups and small and medium enterprises are major sources of jobs at all skill levels and absorb under-utilised labour.

4. Economic growth

- They mobilise savings and investment, create wealth, pay taxes and add to GDP; competition from new entrants raises productivity and gives customers better value.
- Industries set up in less developed regions reduce regional disparities and bring roads, power and other infrastructure.

5. Sustainable development

- Engineers who turn entrepreneurs integrate environmental, social and governance principles into their ventures: eco-friendly products, green supply chains and renewable energy.
- Social entrepreneurs solve problems such as poverty, education, healthcare and climate change through sustainable solutions for marginalised communities.

Entrepreneur	Venture	Contribution
A. G. Bell	Bell Telephone Company	Invented the telephone; created the telecommunications industry.
Steve Jobs and Steve Wozniak	Apple	Smaller, user-friendly, cheaper computers; later phones.
Jeff Bezos	Amazon	Pioneered e-commerce; jobs and a global marketplace.
Larry Page and Sergey Brin	Google	Search engine ranking pages by popularity.

Conclusion: entrepreneurial engineers combine invention with enterprise, advancing technology, creating employment and supporting sustainable, balanced economic growth.

3	Discuss the role of entrepreneurship in promoting innovation and contributing to economic development. Support your answer with real-world examples. Source: Unit I | SEE Feb/Mar 2026, Q2a; CIE-I, 25/11/2024, Q5; SEE Mar/Apr 2023 (18HSI51), Q7a; SEE Jul 2019 Fast Track (16HSI51), Q7c; Question Bank, Unit 1 Topic 1 Q3	08	3	1

Entrepreneurship contributes to economic progress by fostering innovation, improving market dynamics and enhancing the quality of life. Economist Joseph Schumpeter described this as creative destruction, in which old industries are replaced by new innovations.

Role in innovation and economic development

(1) **Driving innovation:** new technologies, products and services create competitive advantage and new markets.
(2) **Employment generation:** start-ups and SMEs are key job creators for varied skill levels.
(3) **Wealth creation and distribution:** mobilising savings and investment generates income that flows through the economy and spreads opportunity across regions.
(4) **Balanced regional development:** industries in backward areas reduce regional disparities; roads, electricity and other infrastructure follow.
(5) **Promoting social change:** social entrepreneurs address poverty, education and healthcare with sustainable solutions.
(6) **Economic dynamism:** diversification reduces dependency on traditional industries, and disruptive business models keep the economy resilient.

Real-world examples

- **Apple and Tesla:** innovation entrepreneurship, creating the iPhone and electric vehicles and whole new product categories and supply chains.
- **Amazon and Facebook:** scalable ventures that created e-commerce and social networking, employing large numbers worldwide.
- **Microfinance institutions and co-operatives:** social enterprises that bring income and credit to marginalised groups.
- **Indian entrepreneurs such as Azim Premji and Lakshmi Mittal:** built large enterprises that create jobs and national wealth.

Conclusion: by turning ideas into enterprises, entrepreneurs make the economy more innovative, more productive and more inclusive.

4	Differentiate between innovative, imitative and Fabian entrepreneurs with suitable examples. Analyse how each type contributes differently to industrial and economic growth. Source: Unit I | SEE Feb/Mar 2026, Q2b; Question Bank, Unit 1 Topic 2 Q2	08	4	1

Entrepreneurs are commonly classified by their approach to innovation and change. The notes list innovative, imitative and Fabian entrepreneurs under types of entrepreneurs; the standard classification (after Clarence Danhof) is summarised below.

Type	Nature	Examples	Contribution
Innovative	Introduces new products, new methods of production or new markets; takes high risks and drives change.	Steve Jobs (Apple), A. G. Bell (telephone), Tesla.	Creates new industries and technological progress; the main source of economic growth and structural change.
Imitative (adoptive)	Does not invent but adopts and adapts successful innovations of others, often with local improvements.	Firms that copy proven models, such as local online marketplaces following the Amazon model.	Spreads technology, widens competition, creates jobs and builds industry quickly, especially in developing economies.
Fabian	Cautious and sceptical; adopts a change only when convinced it is necessary or when delay would mean loss.	A traditional family firm that digitalises only after competitors do.	Gives stability and continuity, but contributes little to innovation and grows slowly.

A fourth type, the **drone entrepreneur**, refuses to change even when circumstances demand it and risks being driven out of the market.

Differing contributions

- Innovative entrepreneurs expand the frontier of what is possible and create the demand that others serve.
- Imitative entrepreneurs take the innovation to the mass market, lowering prices and creating employment.
- Fabian entrepreneurs preserve existing enterprises and traditions and avoid the risk of failed experiments, but are first to lose ground when markets change.

Note: The notes mention these three types only in the syllabus line; the definitions and examples above are from the standard classification.

5	What do you understand by the concept of entrepreneurship? Explain the characteristics of a successful entrepreneur. Source: Unit I | SEE Apr 2024 (18HSI51), Q7a; SEE Nov/Dec 2019 (16HSI51), Q7a; SEE Jan/Feb 2021 (18HSI51), Q7b; SEE Jan/Feb 2021 (16HSI51), Q7a; SEE Jul 2019 Fast Track (16HSI51), Q7b; SEE Nov/Dec 2018 (16HSI51), Q7a	08	2	1

Concept of entrepreneurship

Entrepreneurship is the process of identifying an unmet need or problem, creating a solution and organising resources to deliver value. It includes opportunity recognition, resource mobilisation and risk management. The entrepreneur, from the French entreprendre (to undertake), is an innovator, creator, organiser and risk-bearer, and the activity is full of creativity: new products, processes, markets or sources of supply.

Characteristics of a successful entrepreneur

(1) **Self-motivation and discipline:** drives himself to work even when there is no immediate pay cheque or supervisor.
(2) **Vision:** foresees opportunities and charts a strategic path; sets SMART (specific, measurable, achievable, realistic, time-bound) goals.
(3) **Risk tolerance:** takes calculated risks and accepts uncertainty.
(4) **Innovation, creativity and curiosity:** finds creative solutions with limited resources and keeps learning.
(5) **Leadership and team building:** inspires and guides a team towards business objectives.
(6) **Understanding of what he offers:** knows the product or service and how it is positioned in the market.
(7) **Networking and communication:** builds relationships and recognises partnership opportunities.
(8) **Basic money management:** understands how money works to run the business on sound principles.
(9) **Flexibility and adaptability:** adapts to changing markets, processes and customer needs.

(10) **Passion and honesty:** passion sustains motivation and resilience; honesty and integrity build reputation, trust and finance.
(11) **Planning and emotional intelligence:** has backup plans and manages relationships and failure with resilience.

Conclusion: success depends on a combination of such traits and skills, most of which can be developed through education and experience.

6	Create a case study on how perseverance as a characteristic helped a famous entrepreneur achieve success. (Write a note on any one successful entrepreneur.) Source: Unit I | CIE-I, 25/11/2024, Q4; SEE Nov/Dec 2019 (16HSI51), Q7b; SEE Nov/Dec 2018 (16HSI51), Q7a	10	3	1

Case: Walt Disney, from bankruptcy to the Disney studios

Background

Walter Elias Disney spent his younger years in Chicago drawing and painting pictures and selling them to friends and neighbours. He took formal classes in drawing and photography, worked as a cartoonist for a local newspaper, made commercials using cut-out animation, and even served as a Red Cross ambulance driver in France.

The setbacks

He set up a small animation business and released cartoon series such as Laugh-O-Grams and Alice in Cartoonland. The business did not last: the studio could not repay its debts and he had to declare bankruptcy.

How perseverance helped

- **He treated failure as a cue, not an end:** he moved to Hollywood with his brother and set up the Disney Brothers' Studio.
- **He kept going through early snags:** the new studio met problems at first, but he continued to create.
- **He kept improving his ideas:** the creation of Mickey Mouse made everything take off.
- **Passion for the work:** the same persistence later built the Disney film studio and the Disneyland theme parks.

Outcome

From a bankrupt animator he became one of the best-known creators of entertainment, and his name is now a global brand.

A supporting example: Steve Jobs

Jobs founded Apple at 21 in a family garage with Steve Wozniak. After the early successes of the Apple I and II, later releases did not do well, IBM took the lion's share of the market and he was edged out of Apple in 1985. He set up his own ventures, including Pixar, returned to Apple as CEO in 1997 and started its golden age.

Lessons

- Persistence turns a failure into learning and a new start.
- Perseverance is not stubbornness: both entrepreneurs changed their approach and place, and knew when to move to a new idea.
- Success usually comes after repeated attempts, not on the first try.

7	Illustrate with appropriate examples the myths of entrepreneurship. Source: Unit I | SEE Apr/May 2024 (21HS51A), Q7b; SEE Nov/Dec 2019 (16HSI51), Q7b	08	3	3

A myth is a popular belief about entrepreneurship that does not hold in practice.

No.	Myth	Reality, with example
1	Entrepreneurs are born that way.	Entrepreneurship is a learned skill; entrepreneurs may be extroverts or introverts, big-picture or detail-focused.
2	The only requirement is a good idea.	Even the best ideas need execution, planning, talent, leadership and communication.
3	Starting a business guarantees freedom.	You leave an old schedule but gain new demands; the work does not end at 5 p.m.

4	Launching a company quickly leads to wealth.	Some succeed immediately, others take time; timing expansion and sustaining growth are the real tasks.
5	Only money motivates employees.	Many would rather earn less at a job they enjoy; the entrepreneur must build a culture that engages people.
6	Businesses either flourish or fail.	Patience and strategy matter; some firms that falter at first go on to healthy growth.
7	All responsibility falls on the entrepreneur.	Doing everything alone leads to burnout; collaboration and delegation keep the firm healthy.
8	There is a silver bullet key to success.	Success needs a combination of ideas, people and resources at the right time and place, plus hard work and past failures.
9	Businesses need someone with an MBA at the helm.	A degree has value but is not required; many founders are engineers using technical knowledge, and some have no degree.
10	Quitting is for losers.	Not every idea becomes a sustainable company; successful entrepreneurs try many ideas, and knowing when to walk away is critical.

Other commonly held myths: entrepreneurs are motivated only by profit, entrepreneurship needs a revolutionary idea (incremental innovations also succeed), and failure means the end (it is a learning opportunity).

8	List and explain the characteristics of the different entrepreneurial styles. Source: Unit I | SEE Nov/Dec 2019 (16HSI51), Q8a	08	–	–

Entrepreneurial (leadership) style is the way the entrepreneur directs people and takes decisions. Six styles are described below.

Style	Characteristics	Works best when	Limitation
Laissez-faire	Gives team members great freedom over how they work and their deadlines; feedback only when needed.	The team is highly trained, skilled and self-responsible.	Output suffers if the team cannot manage time or is not self-motivated.
Autocratic	The leader decides without consulting the team and imposes decisions.	Quick decisions are needed and team input is not essential.	One-way approach can demoralise the team.
Transactional	Management and team set tasks together; rewards for completion, penalties for falling short.	The environment is incentive-driven and clarity is needed.	Can seem amoral and does not suit knowledge-based or creative work.
Democratic (participative)	Values inputs and involves members in decisions, though the leader makes the final decision.	Creativity and engagement are needed.	Slow; unsuited to a crisis needing quick decisions.
Pacesetting	Sets high performance standards for self and team and leads by example.	The team is already motivated and skilled.	Overwhelms people and lowers morale of those who cannot keep up.
Transformational	Shares a vision, communicates effectively, inspires and expects the best while showing accountability.	Long-term change and high engagement are the goals.	Depends heavily on the leader's qualities.

No single style is best; effective entrepreneurs choose the style to suit the team, the task and the situation.

9	Analyse the different entrepreneurial development models. Source: Unit I | SEE Apr/May 2024 (21HS51A), Q7a	08	3	4

Entrepreneurial development models describe the ways in which entrepreneurial ventures are organised and grown.

Model	Description	Capital	Examples
Large business entrepreneurship	Massive firms that sustain growth through innovative products and services, with constant R&D to meet changing customer needs.	Internal funds and the capital market.	Apple, Samsung.
Small business entrepreneurship	Starts with one person, a basic idea, passion and determination; about 75 per cent of businesses follow this model; hires local manpower and family members.	Friends and family.	Coffee shop, hair salon, grocery store, consultant.

Social entrepreneurship	Aims to change the world through innovation; focus on a social cause rather than profit; includes non-profits.	Grants, donations, surplus.	Microfinance institutions, co-operatives, educational and welfare programmes.
Scalable start-up entrepreneurship	Confident of a vision that can change the world; grows fast after the first success.	Venture capital.	Facebook.
Innovation entrepreneurship	Finds gaps in the market and uses innovative thinking and technology to build products that make life easier.	Own funds, investors.	iPhone, Tesla.

Analysis

- The models differ mainly in **purpose** (profit, growth or social benefit), **scale** and **source of funds**.
- Large business and innovation models depend on R&D, scalable start-ups on venture capital and fast growth, small businesses on personal commitment and local resources, and social ventures on mission over profit.
- A start-up can move from one model to another: a small innovative venture may become scalable with venture capital and eventually a large business.

10	Determine the measures required for capacity building for entrepreneurship in India. Source: Unit I | SEE Apr/May 2024 (21HS51A), Q8b	08	3	3

Capacity building means strengthening the knowledge, skills, finance and environment available to entrepreneurs. It is needed because entrepreneurs face problems such as limited access to capital, intense competition, regulatory barriers, lack of mentorship, weak risk management and inadequate market research.

Measures

(1) **Education and training:** comprehensive training in entrepreneurship, business planning, financial management and marketing.
(2) **Access to finance:** affordable and accessible financing options to fund ventures.
(3) **Mentorship programmes:** link experienced entrepreneurs with beginners for knowledge transfer and guidance.
(4) **Networking opportunities:** events and platforms for collaboration and shared learning.
(5) **Incubators and accelerators:** provide resources, mentorship and a conducive environment for growth.
(6) **Government policies:** simplify regulation and offer incentives for an enabling environment.
(7) **Technology adoption:** use of technology in business processes to streamline operations, reach wider markets and stay competitive.

Conclusion: collaboration among government, the private sector and educational institutions creates a supportive ecosystem that empowers entrepreneurs, spurs innovation and contributes to economic development.

11	Ms. **Ananya Rao**, a young engineer working in a technology firm, is passionate about developing sustainable energy solutions. While her company supports innovation internally, she is uncertain whether to continue as an intrapreneur within the organisation or to start her own venture focused on solar-powered charging systems for rural areas. She discusses her ideas with colleagues who share her vision, and together they plan to explore market opportunities and assess risks. However, her family believes entrepreneurship is too risky and that only people with large capital or business backgrounds can succeed. Based on the above scenario, analyse the following issues: (i) Differentiate between an entrepreneur and an intrapreneur in the context of Ananya's situation. (ii) Discuss the emerging trends in entrepreneurship that could influence her decision to start a new venture. (iii) Evaluate the characteristics of a successful entrepreneur that Ananya and her team should develop to ensure business success. (iv) Identify and explain the myths about entrepreneurship reflected in the case and how they can be overcome. (v) Explain the role of entrepreneurial teams in fostering innovation and sustaining growth in start-ups like Ananya's. Source: Unit I | CIE-I, 07/11/2025	10	4	1

(i) Entrepreneur and intrapreneur in Ananya's situation

Aspect	If she starts her own venture (entrepreneur)	If she stays in her company (intrapreneur)
Risk	Bears personal financial and business risk.	The organisation bears the risk.
Resources	Must build resources and funding independently.	Uses the firm's resources and R&D.
Autonomy	Operates independently.	Works within organisational policies.
Reward	Keeps all profits (and losses) of the solar charging business.	Shares rewards with the company.

(ii) Emerging trends

- Technology-driven entrepreneurship (AI, IoT, digital platforms) and agile business models.
- Social and sustainable entrepreneurship with ESG principles: directly relevant to solar energy for rural areas.
- Alternative funding such as crowdfunding, angel investing and venture capital; remote work and virtual teams; globalisation and access to new markets; growing diversity, with women and youth entrepreneurs.

(iii) Characteristics to develop

Vision (a clear plan for rural solar charging), risk tolerance (calculated, not reckless), innovation, leadership, self-motivation, flexibility, networking and money management; and passion aligned with values (sustainability), which gives resilience and attracts talent and customers.

(iv) Myths in the case and how to overcome them

- **Only people with large capital or a business background succeed:** an MBA or business degree is not required (many founders are engineers using technical knowledge), and entrepreneurs are made, not born. Overcome by training, mentors, incubators and alternative finance.
- **It is too risky (businesses either flourish or fail):** success needs patience and strategy; risk can be managed through market research, pilots, backup plans and calculated risk-taking.
- **All responsibility falls on the entrepreneur:** Ananya's plan to work with colleagues shows the remedy, which is teamwork and delegation.

(v) Role of entrepreneurial teams

- **Skill diversification and collaborative problem-solving:** engineers, marketers and finance people bring varied expertise and synergy.
- **Shared workload and scaling:** the team handles resource management and market expansion as the venture grows and adapts strategy.

- **Investor confidence:** investors give priority to strong, cohesive teams with clear goals, trust and open communication; conflicts must be managed so personal aspirations align with the venture's objectives.

Conclusion: Ananya can start the venture with a committed team, learn the required traits and use sustainability trends and mentors to manage the risk.

12	List and explain the advantages of an E-Cell. Source: Unit I | SEE Nov/Dec 2018 (16HSI51), Q7b	06	–	–

An **E-Cell (Entrepreneurship Cell)** is a student-run body on a campus that promotes an entrepreneurial culture. The E-Cell of IIT Bombay, for example, was formed in October 1998 and runs entirely by students.

(1) **Bridge between stakeholders:** it connects students, staff, research labs, the institute's innovation centre and alumni.
(2) **Idea spotting and execution:** it helps students spot business ideas and execute them successfully.
(3) **Networking with the ecosystem:** events and summits bring students together with entrepreneurs, mentors, angel investors, venture capital firms and corporates.
(4) **Competitions:** business plan competitions such as Eureka! give practice in planning and pitching.
(5) **Exposure to trends:** panel discussions on topics such as social entrepreneurship and job versus entrepreneurship, and a start-up expo, showcase current trends and success stories.
(6) **Talent platform:** an online platform for start-ups to hire freelancers, interns or co-founders.

13	Explain the meaning and definition of entrepreneurship. Distinguish between an entrepreneur, a manager and an innovator. Illustrate your answer with an example of a successful entrepreneur in India. Source: Unit I | Question Bank, Unit 1 Topic 1 Q1	10	3	1

Meaning and definition

Entrepreneurship is the process of identifying an unmet need or problem, creating a solution and organising resources to deliver value, including opportunity recognition, resource mobilisation and risk management. The entrepreneur is an individual who identifies opportunities, assumes risk and creates value by launching and managing a business.

- **Schumpeter:** the entrepreneur is an innovator who introduces new combinations (a new product, method of production, market or source of supply) and causes creative destruction of old industries.
- **Drucker:** the entrepreneur always searches for change, responds to it and exploits it as an opportunity.

Characteristics

Risk-taking, innovation, leadership and opportunity recognition, together with vision, self-motivation, flexibility and networking.

Entrepreneur, manager and innovator

Basis	Entrepreneur	Manager	Innovator
Role	Identifies the opportunity and creates and owns the venture.	Plans, organises and controls the operations of an existing enterprise.	Creates new ideas, products or methods.
Risk	Bears the financial and business risk.	Salaried; the owner bears the risk.	May not bear commercial risk.
Outcome	A new business that creates value and profit.	Efficient achievement of set goals.	An invention that may or may not be commercialised.

An entrepreneur often combines the innovator's idea and the manager's skills, which turns an invention into a business.

Example: N. R. Narayana Murthy

Narayana Murthy co-founded Infosys in 1981 with a small group of colleagues. He spotted the opportunity in software services, took the risk of leaving a salaried career, and built the firm into a global business that created thousands of jobs and earned export revenue for India. The notes also name Azim Premji, Lakshmi Mittal and Ekta Kapoor as famous Indian entrepreneurs.

Note: The notes do not compare entrepreneur, manager and innovator; that table and the Indian example are from standard sources.

14	Describe the scope of entrepreneurship across different sectors. How can entrepreneurial opportunities differ between urban and rural settings? Provide suitable examples. Source: Unit I | Question Bank, Unit 1 Topic 1 Q2	10	3	1

Scope of entrepreneurship

(1) **Commercial:** ventures that produce and trade goods and services for profit (small businesses, start-ups, large firms).
(2) **Social:** ventures for a social cause rather than profit, such as microfinance institutions, co-operatives, education and welfare services.
(3) **Agricultural:** agri-processing, agri-delivery, cold storage and allied activities.
(4) **Technological:** technology-driven ventures using AI, IoT, blockchain and digital platforms.
(5) **Corporate:** intrapreneurship inside established firms, and large-business entrepreneurship such as Apple and Samsung.

Role in sectors

- **Manufacturing:** new products and processes, jobs and industrial development.
- **Services:** banking, e-commerce, logistics, health and education; the largest scope for new ventures.
- **Agriculture:** better value for farm produce, local processing and rural employment.

Urban and rural opportunities

Aspect	Urban	Rural
Opportunities	Technology, services, platforms, high demand and dense customers.	Agriculture, agro-processing, handicrafts, co-operatives, basic services and energy.
Resources	Capital, skills and infrastructure easy to reach.	Local raw materials and labour; weaker capital and infrastructure.
Example	Zomato (online food delivery).	Amul (dairy co-operative).

Balanced growth needs both: rural enterprises reduce regional disparity, while urban ones drive innovation and scale.

15	GreenFleet Logistics, founded by an engineer, offers last-mile delivery using electric vehicles in Tier-2 cities. The founder aims to expand to agri-delivery and cold storage sectors. What does this case show about the scope of entrepreneurship in modern economies? Source: Unit I | Question Bank, Unit 1 Topic 1 Q4 (case study)	10	4	1

GreenFleet Logistics shows that entrepreneurship in a modern economy is not confined to one sector, technology or place. Its scope has four dimensions.

(1) **Sectoral scope:** the venture integrates transport, agriculture and logistics, moving from last-mile delivery to agri-delivery and cold storage.
(2) **Technological scope:** it uses electric vehicles, IoT tracking and data analytics; an engineer-founder converts technology into a service.
(3) **Geographical scope:** it serves Tier-2 cities and expands beyond metros towards rural areas, reducing regional disparity.
(4) **Social and environmental scope:** electric delivery reduces the carbon footprint and supports sustainability; the venture also creates local jobs.

Conclusion: modern entrepreneurship crosses sectors, uses technology, reaches new regions and serves social goals alongside profit.

16	In Rajasthan, a young graduate named Sneha started an e-commerce platform selling handmade pottery. Local artisans joined her co-operative, earning three times more than before. How does Sneha's initiative demonstrate the importance of entrepreneurship in regional development? Source: Unit I | Question Bank, Unit 1 Topic 1 Q5 (case study)	06	4	1

Sneha's e-commerce platform for handmade pottery shows the importance of entrepreneurship in regional development.

- **Job creation:** she engaged about 70 rural artisans in her co-operative.
- **Cultural preservation:** dying craft forms were revived.
- **Economic impact:** artisans earned about three times more than before, raising local income and the standard of living.
- **Digital inclusion:** artisans were linked to wider markets through the online platform.

This agrees with the roles listed in the notes: employment, wealth creation, balanced regional development and social change.

17	Explain the contributions of any three successful entrepreneurs towards the economic growth and development of their country. How did their innovations or enterprises create employment opportunities and promote industrial or technological advancement? Source: Unit I | Question Bank, Unit 1 Topic 1 Q6	06	–	1

Three entrepreneurs whose enterprises contributed to economic growth and technological progress:

Entrepreneur	Enterprise and innovation	Impact on growth and employment
Steve Jobs	Co-founded Apple at 21 with Steve Wozniak; made computers smaller, more intuitive and cheaper; later the iPhone.	Created whole product categories and an ecosystem of suppliers, software developers and retail jobs; drove advances in computing and mobile technology.
Bill Gates	Co-founded Microsoft in 1975; software for early personal computers.	Built the largest software business; made computing widely usable and created a global software industry with large employment and exports.
Jeff Bezos	Founded Amazon.com in 1995 as an online bookstore.	Pioneered e-commerce; created jobs in logistics, technology and for sellers; opened a global marketplace and raised productivity of trade.

Their ventures contributed to GDP and exports, created employment directly and through supply chains, encouraged start-ups, and led to new technologies, which shows that entrepreneurship drives industrial and technological advancement.

18	Apply suitable techniques to identify potential entrepreneurial opportunities in the field of renewable energy or sustainable products. Analyse how these techniques help in evaluating the feasibility of a new business idea. Source: Unit I | Question Bank, Unit 1 Topic 2 Q1	10	3	1

Opportunity identification techniques turn trends and problems into specific business ideas. Applied to renewable energy and sustainable products:

Technique	Application	What it shows about feasibility
Environmental scanning	Study policy support for solar, falling panel costs, power shortages and demand for green products.	Whether the external environment favours the venture.
Trend analysis	Growing consumer preference for sustainable products and EV and solar adoption.	Size and timing of demand.
Problem identification	Rural households without reliable power: solar charging or cooking appliances.	Real customer need and willingness to pay.
Market research and competitor study	Surveys of price sensitivity; existing solar and eco-product suppliers.	Demand, competition and gaps.
Brainstorming and creativity	Ideas such as solar-powered chargers, recycled-material packaging, biogas units.	Innovation potential and differentiation.
SWOT analysis	Compare the team's technical skills and capital with threats such as subsidy changes.	Fit between idea, resources and risks.

How the techniques help evaluate feasibility

- **Market demand:** research and trends show whether customers need the product at an affordable price.
- **Competition:** a competitor and gap study shows the room for differentiation.
- **Resources:** SWOT and a feasibility matrix compare capital, skills and technology with what is needed.
- **Innovation potential and risk:** creativity tools and scanning show whether the idea is new, scalable and robust to policy and technology changes.

Ideas are then ranked on feasibility and return, and the best is taken forward to a prototype and plan.

Note: The notes do not list these techniques; the answer follows the techniques named in the question bank.

19	Maya, a design student, noticed increasing e-waste on her campus. She researched and found that over 60 per cent of college e-waste could be repurposed. She started a small business creating eco-jewellery and décor from old circuit boards. Which opportunity identification techniques did Maya use? Source: Unit I | Question Bank, Unit 1 Topic 2 Q3 (case study)	06	4	1

Maya used the following techniques to identify her opportunity:

(1) **Observation and problem identification:** she noticed the growing e-waste problem on her campus.
(2) **Trend analysis:** she recognised the rising interest in sustainability among young people.
(3) **Market gap analysis:** she found that few players offered upcycled art and décor, and that over 60 per cent of e-waste could be repurposed.
(4) **Networking:** she partnered with local recyclers for the supply of circuit boards.

She also used creativity to turn waste into eco-jewellery and décor, a venture that is both profitable and sustainable.

20	Three college friends started an e-commerce venture, ShopEase, an online platform to help small local shop owners sell their products digitally in semi-urban areas where big players had limited reach. They faced low digital literacy among vendors, limited funding and competition from larger marketplaces. They built a mobile app in regional languages, offered free onboarding and digital training, and used AI-based recommendations to match local buyers and sellers. Within a year ShopEase partnered with over 2,000 small businesses and expanded to five cities. What key challenges did the team face, and how did their innovative and collaborative approach as e-entrepreneurs help them achieve success? Source: Unit I | Question Bank, Unit 1 Topic 2 Q4 (case study)	06	–	1

Challenges faced

- Low digital literacy among local vendors.
- Limited funding for platform development.
- Competition from larger online marketplaces.

Innovative and collaborative solutions

- A mobile-friendly app in regional languages, which removed the language barrier.
- Free onboarding and digital training of shopkeepers.
- AI-based recommendations matching local buyers and sellers, and partnership with more than 2,000 small businesses.

Outcome

The venture expanded to five cities within a year, empowered local businesses, increased digital participation, improved market reach and won recognition for digital inclusivity.

Concept link

The three friends show that a collaborative team that combines innovation, teamwork and a social purpose can turn limited resources into a business and social transformation.

21	Differentiate with one example the character and trait of a person. Describe the key mindset traits, skills and professional traits of a successful entrepreneur. Source: Unit I | Question Bank, Unit 1 Topic 3 Q1	–	2	1

Character and trait

Character is the set of moral and ethical qualities of a person, shaped by values and choices (what a person is), such as honesty and integrity. A **trait** is a stable personality characteristic that shows in behaviour across situations, such as risk-taking or extroversion. Example: an entrepreneur who refuses to cheat customers shows honest character; one who readily takes bold bets shows the trait of risk tolerance.

Key mindset traits

- **Vision:** foresees opportunities and charts a strategic path (Steve Jobs's computers for everyone).
- **Risk tolerance and willingness to try:** accepts calculated risk and learns from failure.
- **Innovation and creativity:** finds new solutions with limited resources.
- **Passion and self-motivation:** drives himself even without an immediate pay cheque, and keeps going after setbacks.
- **Curiosity:** keeps learning and changes opinion when facts change.

Entrepreneurial skills

- **Opportunity recognition and strategic thinking:** spotting viable ideas and planning.
- **Leadership and decision-making:** guiding a team and acting under uncertainty.
- **Communication and networking:** building relationships with investors, partners and customers.
- **Money management and problem-solving:** understanding finances and finding solutions.

Professional traits

- **Integrity and honesty:** build reputation and trust; lenders and customers rely on the entrepreneur's word.
- **Accountability and discipline:** doing the work even when not in the mood; owning outcomes.
- **Adaptability and flexibility:** adjusting to markets and technology.
- **Time management and planning:** SMART goals and backup plans.

22	Discuss the emerging trends shaping new-age entrepreneurship. How are technology integration, sustainability and global connectivity transforming entrepreneurial practices in the 21st century? Illustrate with suitable examples. Source: Unit I | Question Bank, Unit 1 Topic 3 Q2	10	4	1

Entrepreneurship in the new age is characterised by dynamic shifts influenced by technology, changing consumer behaviour and an evolving global landscape.

Emerging trends

(1) **Technology integration:** AI, advanced tools and data analytics improve efficiency, reach and innovation.
(2) **E-commerce and digital presence:** firms grow through online platforms and serve global, digital-first customers.
(3) **Remote work and collaboration:** virtual teams across borders bring diverse skills.
(4) **Sustainability and social impact:** environmental and social responsibility are part of business strategy.
(5) **Agile business models:** quick response to market change and resilience.
(6) **Access to global markets:** easier cross-border trade and partnerships.
(7) **Innovation and disruption:** constant disruption of traditional industries.
(8) **Customer experience:** customer-centric approaches using technology.
(9) **Diversity and inclusion:** diverse perspectives that foster creativity.

How technology, sustainability and global connectivity transform practice

- **Technology:** products and services built on platforms and AI, with low cost of reaching customers; Zoom for remote collaboration.
- **Sustainability:** Tesla shows how clean technology can become a business; green supply chains and renewable energy.

- **Global connectivity:** Amazon and online marketplaces let even small firms sell worldwide, and funding comes from crowdfunding, angel investors and venture capital.

Conclusion: adaptability and continuous innovation define success in new-age entrepreneurship.

23	Start-up MediAI developed an app using AI chatbots for preliminary health advice, reducing doctor waiting times during the pandemic. How does MediAI reflect emerging trends in entrepreneurship? Source: Unit I | Question Bank, Unit 1 Topic 3 Q3 (case study)	–	–	1

MediAI reflects several emerging trends in entrepreneurship:

- **AI-driven innovation:** automation of preliminary health advice with chatbots.
- **Online service model:** telemedicine and remote diagnosis; digital presence and reduced doctor waiting time.
- **Customer personalisation:** AI-based patient profiling and customer-centred service.
- **Social impact and inclusion:** improved health access in rural areas, and agility in responding to a crisis (the pandemic).

24	Bring out four differences between entrepreneur and entrepreneurship with respect to concept, focus, outcome and nature, with a suitable example for each. Source: Unit I | Question Bank, Unit 1 Topic 3 Q4	04	–	1

An **entrepreneur** is the person who starts and manages a business, while **entrepreneurship** is the process of creating and running the venture.

Basis	Entrepreneur	Entrepreneurship	Example
Concept	An individual.	A process.	Ratan Tata (entrepreneur); creating the Tata businesses (entrepreneurship).
Focus	On the person and his qualities.	On the activity of creating value.	Steve Jobs; launching Apple.
Outcome	A business leader.	A business venture and its benefits to society.	Jeff Bezos; Amazon.
Nature	A qualitative trait set (vision, risk-taking).	A functional process (identify, mobilise, manage).	A founder's passion; the steps to build the firm.

25	Bring out six differences between an entrepreneur and an intrapreneur. Source: Unit I | Question Bank, Unit 1 Topic 4 Q1	06	2	1

Aspect	Entrepreneur	Intrapreneur
Definition	Individual who starts a new venture.	Innovator working within an existing organisation.
Risk	Takes personal financial and business risks.	Risks are borne by the organisation.
Resources	Builds resources independently.	Leverages organisational resources.
Autonomy	Operates independently.	Functions under organisational policies.
Reward	Retains all profits or losses.	Shares rewards with the company.
Example	Steve Jobs founding Apple.	An engineer who leads a new product project inside an established firm.

26	Discuss the importance of teams and team dynamics in entrepreneurship. How do strong entrepreneurial teams contribute to venture growth and investor confidence? Additionally, explain any four common characteristics of successful entrepreneurs with suitable examples. Source: Unit I | Question Bank, Unit 1 Topic 4 Q2	10	4	1

Entrepreneurial ventures are built by a team of people whose combined traits decide success.

Importance of teams

- **Skill diversification:** varied expertise and perspectives (technical, marketing, finance).
- **Collaborative problem-solving:** synergy in decisions and innovation.
- **Shared workload:** delegation avoids burnout of one person.

Team dynamics

Effective teams share clear goals, trust and open communication. Conflicts must be managed, and personal aspirations aligned with the objectives of the venture.

Scaling and investor confidence

- Teams handle the complexity of growth: resource management and market expansion, and adapt strategy over time.
- Investors prioritise strong, cohesive teams because the team strongly affects the venture's success.

Four common characteristics of successful entrepreneurs

(1) **Self-motivation:** pushes forward without supervision or immediate pay.
(2) **Taking calculated risks:** accepts risks that are more likely to pay off.
(3) **Networking:** connects with partners and investors and spots opportunities.
(4) **Passion:** keeps motivation and resilience when discouraged.

Others: understanding the market offering, money management and flexibility. **Conclusion:** teamwork and entrepreneurial traits together lead to long-term business success.

27	Write the characteristics and skills which you liked about your favourite entrepreneur which made him successful. Source: Unit I | Question Bank, Unit 1 Topic 4 Q3	10	2	1

Entrepreneur chosen: Steve Jobs (co-founder of Apple).

Characteristics that made him successful

- **Vision:** to make computing accessible by making computers smaller, more intuitive, user-friendly and cheaper.
- **Passion and creativity:** a lifelong interest in electronics and design; Apple began in a family garage when he was 21.
- **Resilience and perseverance:** he was edged out of Apple in 1985, set up other ventures including Pixar, and returned in 1997 as CEO to start Apple's golden age.
- **Focus on the customer experience and innovation:** products that created new categories.

Skills

Leadership, strategic thinking, product design sense, communication, and the ability to build and inspire teams.

What I liked and learned

That failure and rejection can be turned into a fresh start, that passion for the product matters more than quick profit, and that focus on user experience builds loyal customers.

Note: This is a personal-choice question; the answer is a model using a founder described in the notes.

28	Give examples of two entrepreneurs who brought change and shaped the world according to their vision. Source: Unit I | Question Bank, Unit 1 Topic 4 Q4	10	2	1

Entrepreneurs are creators who introduce something new and change an industry or society according to their vision. Two examples:

1. Steve Jobs (Apple)

- **Background:** founded Apple at 21 with Steve Wozniak in a family garage; left in 1985, founded Pixar, and returned as CEO in 1997.
- **Vision and innovations:** to make computers smaller, more intuitive and cheaper; personal computers (Apple I and II) and later the iPhone with a focus on user experience.
- **Impact:** changed personal computing and mobile technology, and the way people communicate and design products.

2. Jeff Bezos (Amazon)

- **Background:** studied electrical engineering and computer science, became the youngest vice president in a Wall Street investment company, then left to start Amazon.com, which opened on 16 July 1995 as an online bookstore.
- **Vision and innovations:** to pioneer e-commerce and build the largest online marketplace.
- **Impact:** transformed retail and logistics, created jobs, and gave businesses of all sizes access to global customers.

Conclusion: visionary leadership and innovation, together with persistence, drive transformative change.

29	List six common entrepreneurial myths. Source: Unit I | Question Bank, Unit 1 Topic 4 Q5	06	–	1

(1) **Entrepreneurs are born that way:** entrepreneurship is a learned skill.
(2) **The only requirement is a good idea:** execution, planning and leadership are also needed.
(3) **Starting a business guarantees freedom:** it brings new demands and long hours.
(4) **Launching a company quickly leads to wealth:** growth takes time and timing.
(5) **Only money motivates employees:** people also value enjoyable work and culture.
(6) **Businesses either flourish or fail:** many firms recover and grow after a slow start.

Other myths: all responsibility falls on the entrepreneur, there is a silver bullet to success, businesses need someone with an MBA, and quitting is for losers.

30	Discuss the following entrepreneurial myths: (1) Entrepreneurs are born that way. (2) Businesses need someone with an MBA at the helm. Source: Unit I | Question Bank, Unit 1 Topic 4 Q6	06	–	1

Myth 1: Entrepreneurs are born that way

Many assume that only people with natural talent can be entrepreneurs. In fact, almost anyone can become an entrepreneur if they learn the necessary skills; some may adjust to the role faster, but there is no rule that only certain types of people can found companies. Entrepreneurs may be outgoing or quiet, big-picture thinkers or detail-oriented, and entrepreneurship is a learned skill, not a natural ability. Example: Bill Gates and Jeff Bezos built their skills through study and practice.

Myth 2: Businesses need someone with an MBA at the helm

A formal education has value, but the market does not require an MBA or business degree. Many founders are engineers who use deep technical knowledge to identify gaps and devise solutions, and some have no degree at all. Experience, a team and mentors can supply the business skills. Example: Steve Jobs, Bill Gates and Mark Zuckerberg did not complete a degree.

Both myths mislead aspiring entrepreneurs and can be overcome by training and practical learning.

31	A start-up, EcoSmart Solutions, founded by three young engineers (Amit, a technical innovator; Riya, skilled in marketing; Sanjay, experienced in finance and operations), aims to create affordable solar-powered home appliances for rural areas. In the first year the team faced lack of market understanding, financial mismanagement and production delays, and tension over R&D versus early market testing. Investors lost confidence. The team then held regular strategy meetings, defined roles, sought mentorship, used agile project management and delegated tasks; within a year it launched its first solar cooker, secured funding and expanded into three states. What key team-related challenges did EcoSmart face, and how did effective teamwork help them achieve business growth? Source: Unit I | Question Bank, Unit 1 Topic 4 Q7 (case study)	06	–	1

Team-related challenges

- Lack of coordination and conflicting priorities: Amit wanted heavy R&D, Riya wanted early market testing.
- Unclear roles, weak financial control and poor communication, which led to production delays.
- Loss of investor confidence because of slow progress.

Team-based strategies used

- Clearly defined roles for the technical, marketing and finance founders.
- Regular strategy meetings and better communication channels.
- Mentorship from an experienced entrepreneur.
- Agile project management and effective delegation.

Outcome

Efficiency improved, investor confidence returned, the first solar cooker was launched, funding was secured and the firm expanded into three states.

Concept link

The case shows how an entrepreneurial team, with diverse skills, shared goals, trust and conflict management, drives innovation, adaptability and growth, and how investors prefer cohesive teams.

32	"Successful entrepreneurs possess a combination of personal, managerial and behavioural qualities." Discuss the important characteristics and traits of successful entrepreneurs, with suitable examples. Source: Unit I | CIE-1, 09/10/2026, Q4	10	2	1

An **entrepreneur** is an innovator, creator, organiser and risk-bearer who recognises an opportunity and builds resources around it. Success depends on a combination of qualities which can be grouped as personal, managerial and behavioural, and most of them can be developed through education and experience.

Group	Characteristics and traits	Examples
Personal qualities	**Self-confidence and self-motivation:** works hard without a supervisor or an immediate pay cheque. **Risk-taking:** takes calculated risks and accepts uncertainty. **Creativity and innovation:** finds new solutions with limited resources. **Need for achievement:** sets SMART goals and works to reach them. **Persistence and determination:** continues after failure.	Dhirubhai Ambani built Reliance from a small trading business through risk-taking and determination. Steve Jobs showed creativity and innovation at Apple.

Managerial qualities	**Vision and planning:** foresees opportunities, sets a strategic direction and keeps backup plans. **Leadership and team building:** inspires and guides a team towards the objectives. **Decision making and organising:** mobilises people, money and materials. **Understanding of the product and the market:** knows what is offered and how it is positioned. **Basic money management** and **networking:** runs the business on sound financial principles and builds relationships.	N. R. Narayana Murthy led Infosys with a clear vision and a strong team. Kiran Mazumdar-Shaw built Biocon by organising resources and a technical team.
Behavioural qualities	**Honesty and integrity:** builds reputation, trust and the confidence of investors. **Flexibility and adaptability:** adjusts to changing markets, technology and customer needs. **Communication and emotional intelligence:** manages relationships and criticism. **Passion and resilience:** sustains motivation and learns from failure. **Social responsibility:** considers employees, customers and society.	Ratan Tata is known for integrity and ethical leadership in business. Jeff Bezos adapted Amazon from an online bookshop to a wide marketplace as the market changed.

Conclusion

No single quality guarantees success. A successful entrepreneur combines personal drive, managerial ability to organise the venture and behaviour that earns the trust of customers, employees and investors.

33	A technology-based start-up wants to use Artificial Intelligence, digital platforms, social entrepreneurship, sustainable business practices and innovative funding models to establish its business. Explain the emerging trends represented in this case and analyse their impact on entrepreneurship. Source: Unit I | CIE-1, 09/10/2026, Q5	10	2	1

The start-up in the case reflects the main trends that now shape new-age entrepreneurship: technology, digital markets, social purpose, sustainability and new ways of raising money.

Emerging trends represented in the case and their impact

Trend	Meaning and example	Impact on entrepreneurship
Artificial Intelligence and technology integration	AI, data analytics and automation are built into products and operations; for example AI chatbots for health advice or customer support.	Raises efficiency, improves decisions and creates new products; lowers the cost of starting and operating, but needs technical skills and brings risks of data privacy and bias.
Digital platforms and e-commerce	Business is run through online marketplaces, apps and social media; for example Amazon, Zomato and Ola.	Gives low-cost access to customers anywhere, supports remote and virtual teams and global markets, but also invites intense competition and dependence on the platform.
Social entrepreneurship	A venture that solves a social problem and measures its success by social impact as well as profit; for example Grameen Bank in microfinance.	Opens opportunities in health, education, rural development and financial inclusion; attracts impact investors and customers who value purpose, but profit and social aim must be balanced.
Sustainable business practices	Green products, renewable energy, recycling and responsible supply chains are part of the business strategy; for example Tesla in clean technology.	Creates new markets, builds brand trust and meets regulation, though sustainable methods may raise the initial cost.
Innovative funding models	Crowdfunding, angel investors, venture capital, impact investing and bootstrapping instead of only bank loans; for example Milaap and Ketto in India.	Makes it easier for young founders without collateral to raise money and test ideas, but the founder shares ownership and faces investor expectations.

Overall impact on entrepreneurship

- **Lower barriers to entry:** digital tools and new funding make it possible for more people, including those without large capital or business backgrounds, to start ventures.
- **New opportunities and faster innovation:** AI and platform business models disrupt traditional industries and keep competition high, so continuous innovation and agility are needed.
- **Wider purpose of business:** social and environmental responsibility becomes part of strategy, so entrepreneurs measure success by profit, people and planet.
- **New challenges:** the founder needs digital skills, awareness of ethics and regulation, and the ability to manage investors, risk and fast change.

Answer: These trends make entrepreneurship more accessible, technology-driven and purpose-oriented, and reward adaptability and continuous innovation.

34	Intellectual property can be categorised into artistic and commercial as per the nature of the work. Justify. List and explain the different types of intellectual property rights with suitable examples. Source: Unit IV | SEE Apr 2024 (18HSI51), Q2a; SEE Mar/Apr 2023 (18HSI51), Q2a; SEE Nov/Dec 2019 (16HSI51), Q2a; SEE Jan/Feb 2021 (16HSI51), Q2b; SEE Nov/Dec 2018 (16HSI51), Q2a	08	1	1

Intellectual property and intellectual property rights

Intellectual property (IP) covers creations of the mind such as inventions, music, books, paintings and brand names. These are intangible assets with commercial value. The creator or owner is given exclusive rights over the creation, called **intellectual property rights**, so that the owner benefits financially and the work is protected. IPR also protects the moral and material interests of authors, as recognised in Article 27 of the Universal Declaration of Human Rights.

Artistic and commercial categories

- **Artistic works** (books, music, software, paintings) are protected by **copyright**.
- **Commercial (industrial) property**, created and used for industrial or commercial purposes, includes **patents, trademarks, industrial designs and trade secrets**.

Types of intellectual property rights

Type	What it protects	Example	Duration
Copyright	Exclusive right to control publication, distribution and adaptation of literary and artistic works.	Book, music, software, painting.	Life of the author plus about 50 to 100 years; 95 years from publication for anonymous works.
Trademark	A symbol, word, logo, design or image that identifies the source of goods and distinguishes them from others.	The logo on Nike products.	Protected when registered; owner can sue for unauthorised use.
Patent	A new, inventive and industrially applicable invention; the right to prevent others making, using, selling or distributing it.	A new machine, process or composition of matter.	Generally 20 years from filing.
Industrial design	Visual design of objects with aesthetic or ornamental value (shape, colour, pattern), two- or three-dimensional.	A bottle shape or fabric pattern.	10 to 25 years.
Trade secret	Designs, formulas, recipes, processes or ideas that give an economic advantage over competitors.	The formula of a soft drink.	Unlimited, so long as secrecy lasts.

Trade secrets differ from the others because the owner must keep the secret and the right is not backed by registration; once the secret leaks, anyone can use it.

35	Define a patent and explain its salient features that make it a valuable legal protection for inventors. Analyse the key factors determining the patentability of an invention. Source: Unit IV | CIE-I, 07/11/2025; SEE Apr 2024 (18HSI51), Q2b; SEE Mar/Apr 2023 (16HSI51), Q2a	10	3	4

Definition

A **patent** is a right conferred on a person who invents a new machine, process, article of manufacture or composition of matter, giving the exclusive right to prevent others from making, using, selling or distributing the invention without permission. The Indian Patents Act defines an invention as a new product or process involving an inventive step and capable of industrial application.

Salient features

(1) **Novelty:** the invention must be new on the priority date, i.e. not anticipated by publication or use anywhere in the world. For example, an English inventor was denied a patent on an improved ballpoint pen because he had published its details before filing.

(2) **Inventive step:** a technical advance, economic significance or both, that is non-obvious to a person skilled in the art. A tailor adapting cloth to fashion shows skill, not invention; the Tej tube-well strainer, made by a new and cheaper process for local conditions, was held to be inventive.

(3) **Industrial application:** it can be made or used in an industry, covering any useful practical activity as distinct from a purely intellectual or aesthetic one.

(4) **Exclusive right for a limited period:** generally 20 years from the date of filing, kept alive by annual renewal fees.

(5) **Territorial and registered right:** granted by the patent office after examination, and the patentee can assign, license or sue for infringement.

(6) **Disclosure in return for monopoly:** the specification fully describes the invention and the best method; what is not claimed is disclaimed.

(7) **Incremental inventions qualify:** a major breakthrough is not required; small improvements that are new, inventive and industrially applicable can be patented.

Factors determining patentability

Factor	Test
Novelty	Not part of the state of the art; no prior publication, written or oral disclosure or public use before the filing date.
Inventive step	Not obvious to a person skilled in the art; results from independent thought, ingenuity and research.
Industrial application	Utility; capable of being made or used in industry.
Subject matter	Must not fall in the excluded categories (for example atomic energy, mere discovery, mathematical methods and computer programmes per se, methods of agriculture, medical treatment).
Applicant	The true and first inventor, his assignee or legal representative.
Disclosure	A complete specification that describes the invention sufficiently, with claims supported by the disclosure.

36	Discuss in detail the concept, scope and salient features of patents. Explain with examples the kinds of inventions that are patentable under the Indian patent law. How does understanding these distinctions help in promoting innovation? Source: Unit IV | SEE Feb/Mar 2026, Q7a; Question Bank, Unit 4 Part B Q2	08	3	4

Concept and scope

A patent gives the inventor a monopoly for a limited period, generally 20 years from filing, in return for public disclosure of the invention. The patentee can stop others from making, using, offering for sale, selling or importing a patented product, or the product obtained directly by a patented process, in India. Under the TRIPS Agreement, Article 27, patents are available for any inventions, products or processes, in all fields of technology, subject to exclusions; members must also protect micro-organisms and non-biological and micro-biological processes and protect plant varieties by patents or an effective sui generis system.

Salient features

The invention must be **new** (not in the prior art), involve an **inventive step** (non-obvious to a skilled person) and be capable of **industrial application**. Small, incremental improvements qualify if they meet these tests.

Patentable inventions, with examples

- A new machine, device or composition, such as the telephone.
- A new process or an improvement in a process: for example, a method of drying a substance without damage to desired characteristics and with better shelf-life or stability.
- A genetically engineered micro-organism that did not exist in nature, such as the oil-spill degrading bacterium of Dr. Ananda Chakraborty (granted in the United States in 1980), which satisfied novelty, non-obviousness and utility.
- A computer system that is programmed to operate in a new way to produce a required result (the computer along with the programme).
- A new combination of known processes that gives a new result at lower cost, such as the Tej tube-well strainer.

Not patentable: atomic-energy inventions, discoveries of natural substances or scientific principles, mathematical and business methods, computer programmes per se, methods of agriculture and horticulture, medical treatment methods, and plants and animals other than micro-organisms.

Promoting innovation

- Clear rules tell inventors what to protect, so they invest in real technical advances rather than mere discoveries.
- Excluding subject matter such as discoveries, abstract ideas and methods of treatment keeps basic knowledge and public health free for all.
- Disclosure in the specification spreads technology, and after 20 years the invention enters the public domain.

37	Explain the inventions which are not patentable under the Indian Patents Act. Source: Unit IV | SEE Feb/Mar 2025, Q7b; SEE Apr 2024 (18HSI51), Q3a; Question Bank, Unit 4 Part B Q6	08	5	4

The Indian Patents Act, in line with the TRIPS Agreement, lists what are not inventions and cannot be patented. The exclusions fall into two categories: subject matter that cannot technically be called an invention, and inventions denied protection on public policy grounds.

No.	Not patentable	Reason or example
1	Inventions relating to atomic energy.	Central Government has sole responsibility for atomic energy.
2	Inventions whose use would be contrary to public order or morality, or seriously prejudicial to human, animal or plant life, health or the environment.	A new type of gambling machine.
3	Frivolous inventions, or those contrary to established natural laws.	A perpetual motion machine.
4	Mere discovery of a scientific principle or formulation of an abstract theory.	A discovery only unveils a hidden thing; it does not make it useful.
5	Mere new form of a known substance with no enhanced efficacy, new property or new use of a known substance, mere use of a known process, machine or apparatus.	Salts, esters, ethers, polymorphs, isomers and other derivatives are the same substance unless they differ significantly in efficacy.

6	Discovery of living or non-living substances occurring in nature.	A naturally occurring plant or mineral.
7	Substance obtained by mere admixture of components, or the process of producing the mixture.	Simple blending without a new result.
8	Mere arrangement, rearrangement or duplication of known devices working independently in a known way.	Joining two known devices without changing the end result.
9	An invention which in effect is traditional knowledge.	Medicinal uses of turmeric known for centuries.
10	Mathematical or business method, computer programme per se, or algorithms.	Protected, if at all, by copyright.
11	Mere scheme, rule or method of performing a mental act or playing a game.	Rules of a game.
12	Method of agriculture or horticulture.	Policy exclusion.
13	Medicinal, surgical, curative, prophylactic, diagnostic, therapeutic or other treatment of human beings and animals.	Policy exclusion for health.
14	Plants and animals in whole or part, other than micro-organisms, including seeds, varieties and species, and essentially biological processes for producing them.	Plant varieties need a separate (sui generis) system.
15	Literary, dramatic, musical or artistic works including cinematographic works and television productions.	Covered by copyright.
16	Topography of integrated circuits.	A separate field of IP.

Reasons for the exclusions

- **Public interest and policy:** inventions contrary to public order or morality, atomic energy, medical treatment methods and agriculture are kept free of private monopoly.
- **No technical contribution:** discoveries, abstract theories, mathematical and business methods and mere arrangements of known devices add no new technical solution.
- **Other IP regimes cover them:** artistic works, computer programmes and integrated-circuit layouts are protected by copyright or separate laws, and plants by a sui generis system.
- **Protection of knowledge already public:** traditional knowledge and natural substances belong to everybody, and patents on them would be monopolies over existing knowledge.

38	A software company develops an AI tool for legal research and applies for a patent in India. Discuss the challenges it might face under Indian patent law. Source: Unit IV | CIE-I, 25/11/2024	05	3	3

The company will face the following challenges under Indian patent law.

(1) **Subject-matter exclusion:** a computer programme per se, an algorithm and a mathematical or business method are not inventions and cannot be patented. Under Indian law programmes are protected by the Copyright Act, 1957. An AI tool for legal research is largely software and algorithms.

(2) **Showing a technical contribution:** a patent is possible only if the tool is claimed as a computer system programmed to operate in a new way to produce a required result. Claims that only automate searching of legal text, or a mere scheme, rule or mental act, will be refused.

(3) **Novelty and inventive step:** the tool must not be anticipated by any prior publication or use anywhere in the world, and the technical advance must not be obvious to a skilled person; many legal search methods already exist.

(4) **Industrial application:** the claimed result must be a useful practical activity and not a purely intellectual one.

(5) **Sufficient disclosure:** the specification must describe the invention fully with the best method, and the claims must be supported by the disclosure; revealing too little of the model invites refusal or opposition.

(6) **Inventorship and ownership:** the application must name the true and first inventor, and the employer needs an assignment from the employee developers.

(7) **Examination and opposition:** the application is examined for anticipation by prior art and may face pre-grant or post-grant opposition.

Advice: protect the code by copyright, and draft patent claims around the technical system and its effect.

39	Explain the difference between a product patent and a process patent with examples. Source: Unit IV | CIE-I, 25/11/2024	05	1	4

Aspect	Product patent	Process patent
Protects	The product itself: a machine, article of manufacture or composition of matter.	The method or process of making a product or achieving a result.
Right of the patentee	Prevent third parties from making, using, offering for sale, selling or importing the product, by any process.	Prevent third parties from using the process, and from using, offering for sale, selling or importing the product obtained directly by that process in India.
Scope	Wider; covers the product however made.	Narrower; others may make the same product by a different process.
Detection of infringement	Easy: the product can be seen in the market.	Harder: the process is inside the factory.

Examples

- **Product patent:** a new composition such as a new drug compound, or the genetically engineered bacterium able to degrade oil, which the court treated as a product.
- **Process patent:** a new method of manufacturing the strainer (Tej) at lower cost, or a new method of drying a substance that improves its shelf-life.

A claim may cover a process, product, machine or composition of matter, but not the function or result alone.

40	Enumerate the procedure for obtaining a patent in India. Source: Unit IV | SEE Mar/Apr 2023 (16HSI51), Q2b; SEE Nov/Dec 2019 (16HSI51), Q2b; Question Bank, Unit 4 Part B Q8	08	–	–

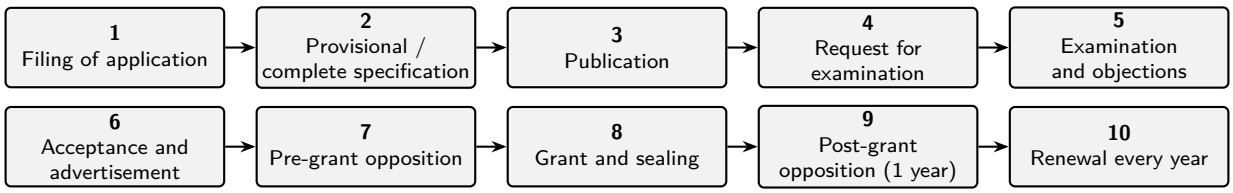


Figure 1: Procedure for obtaining a patent in India

(1) **Who can apply:** the true and first inventor, his assignee, or the legal representative of a deceased inventor. For an employee-inventor the ownership depends on the employment contract.

(2) **Filing:** one application for one invention, in the prescribed form with fees, in the appropriate patent office (by residence, domicile, place of business or place of invention). It states possession of the invention and names the inventors and assignee, and includes a claim of priority where applicable.

(3) **Specification:** a provisional specification (nature of the invention) secures priority, and the complete specification must follow within 12 months or the application is deemed abandoned. A convention application needs a complete specification.

(4) **Publication:** after a prescribed period the application is published with date, number, applicant and abstract (earlier on request); defence-relevant inventions may be kept secret by direction of the Controller. From publication to grant the applicant has the rights of a patentee but cannot sue until the patent is granted.

(5) **Request for examination:** within the prescribed period; otherwise the application is treated as withdrawn.

(6) **Examination:** an examiner checks adequacy of description, title, claims, conformity with the Act, lawful objections and anticipation by prior publication or prior claim. The report is confidential. Objections are communicated; if removed within the prescribed period the specification is accepted, otherwise the application is refused after a hearing.

(7) **Acceptance and advertisement:** the acceptance is advertised in the Official Gazette and the documents are open to public inspection.

(8) **Opposition:** pre-grant opposition by any person after publication, and post-grant opposition within one year of publication of grant.

(9) **Grant and sealing:** the patent is granted, sealed with the patent office seal and entered in the Register of Patents; the patent date is the date of filing.

(10) **Renewal:** annual renewal fees must be paid for the 20-year term.

41	What details should a complete specification contain? Source: Unit IV | SEE Apr/May 2024 (21HS51A), Q2a	08	1	1

The specification is a complete and accurate description of how the invention can be carried out by the method best known to the applicant, ending with claims that define the scope of protection. The complete specification must contain:

(1) A **title** that sufficiently indicates the subject matter of the invention.
(2) Relevant **drawings**.
(3) A **full and particular description** of the invention.
(4) Details of its **operation or use** and the method by which it is to be performed.
(5) Disclosure of the **best method** of performing the invention.
(6) **Claims** defining the scope of the invention, substantiated by the disclosure.
(7) An **abstract** providing technical information on the invention.
(8) A **declaration as to the inventorship** of the invention.

Related points

- The claims relate to a single invention or a group forming a single inventive concept; what is not claimed is disclaimed.
- The complete specification fixes the scope of the patent and is read as a whole (title, body and claims). Claims must be fairly based on the matter disclosed, and sufficient experimental data helps to draft broader claims.
- It can be amended only by disclaimer, correction or explanation, and not so as to claim or describe matter not disclosed in the original.
- An improvement of a patented invention may be granted as a patent of addition, if the patentee is the same.

42	Discuss the grounds on which the conferment or grant of a patent can be opposed. Source: Unit IV | SEE Jul 2019 Fast Track (16HSI51), Q2a	10	–	–

The Indian Patents Act provides for both pre-grant and post-grant opposition.

Pre-grant opposition

After the application is published but before the patent is granted, **any person** may represent in writing to the Controller on the following grounds (the representor does not become a party to the proceedings):

(1) The invention formed part of the prior art on the date of application and was anticipated.
(2) The specification does not clearly and sufficiently describe the invention or the method of performing it.
(3) The applicant has not disclosed the necessary information or has furnished false information in a material particular.
(4) In a convention application, the application was not made within 12 months of the first application.
(5) The complete specification does not disclose, or wrongly mentions, the source or geographical origin of biological material used.

Post-grant opposition

Any interested person may give notice to the Controller **within one year** from the date of publication of the grant on these grounds:

(1) The invention was wrongfully obtained by the applicant.
(2) The invention was anticipated by an earlier patent or other publication.
(3) It was publicly known or used in India before the priority date.
(4) It is obvious and involves no inventive step.
(5) The specification does not sufficiently and clearly describe the invention.
(6) The subject of the patent is not an invention within the meaning of the Act.
(7) Information on corresponding foreign applications furnished to the Controller is false.
(8) The geographical origin of biological material is not disclosed or is falsely disclosed.
(9) In a convention application, the application was not made in the prescribed time.
(10) The invention was anticipated by knowledge, oral or otherwise, available in a local or indigenous community in India or elsewhere.

No other ground is entertained.

Procedure

On receipt of the notice, the Controller informs the patentee and constitutes an **Opposition Board**, which examines the notice and documents and submits recommendations. After hearing the patentee and the opponent, the Controller orders the patent to be **maintained, amended or revoked**.

43	Define patent transfer. Discuss the various methods by which patent rights can be transferred. Source: Unit IV | SEE Feb/Mar 2026, Q7b; SEE Jan/Feb 2021 (16HSI51), Q2a; Question Bank, Unit 4 Part B Q10, Q12	08	3	4

Patent transfer is the passing of the rights in a patent from the patentee to another person. A patent is the exclusive property of the inventor and can be transferred by assignment, grant of licence or operation of law. An assignment, licence or any other interest must be in writing, with the terms and conditions clearly specified, and the person acquiring the title must apply in writing to the Controller to register it in the Register of Patents.

1. Assignment

Transfer of the interest in the patent, wholly or in part, valid over the whole of India or a part of it; the transferor is the assignor and the transferee the assignee.

- **Legal assignment:** through a duly registered agreement; the assignee's name is entered in the Register as proprietor and has all the rights conferred by the assignor.
- **Equitable assignment:** the patentee agrees by a document, such as a letter, to give defined rights with immediate effect; it cannot be registered but can be converted into a legal assignment by a registered written document.
- **Mortgage:** transfer of the rights, wholly or partly, to secure a sum of money; the patentee can get the patent back on refunding the consideration.

2. Licence

A licence gives another person the privilege to make, use or exercise the invention under an agreement. It transfers a right and not an interest in the patent.

- **Voluntary licence:** by written agreement on agreed terms; the Controller and Central Government have no role.
- **Statutory (compulsory) licence:** granted by the Controller or Central Government irrespective of the patentee's will.
- **Exclusive licence:** exclusive right to make, use, sell or distribute to one person to the exclusion of all others, including the patentee; the licensee can sue infringers. Restrictions in licences, such as tying the purchase of other articles, must not be against public interest.

3. Operation of law

On the death of the patentee the interest passes to his legal representative; the Central Government can also acquire a patent for a public purpose.

44	Differentiate between the following: (i) compulsory licensing and voluntary licensing; (ii) legal assignment and equitable assignment; (iii) provisional application and complete application. Source: Unit IV | SEE Nov/Dec 2018 (16HSI51), Q2b	06	–	–

(i) Compulsory licensing and voluntary licensing

Aspect	Compulsory (statutory) licence	Voluntary licence
Granted by	The Controller or the Central Government.	The patentee by written agreement.
Consent	Without the patentee's will.	With the patentee's consent, on agreed terms.
Grounds	Public needs unmet, unaffordable price, not worked in India, national emergency.	Commercial decision of the patentee.

(ii) Legal assignment and equitable assignment

Aspect	Legal assignment	Equitable assignment
Document	Duly registered agreement.	A document such as a letter, not an agreement.
Register	Assignee entered as proprietor.	Cannot be registered; convertible to legal assignment.

(iii) Provisional application and complete application

Aspect	Provisional specification	Complete specification
Content	Only the nature of the invention at the stage of conception.	Full description, best method, drawings and claims.
Effect	Secures the priority date.	Defines the scope of protection; needed for examination.
Time	Filed first.	Must be filed within 12 months or the application is deemed abandoned.

45	Can any invention be patented after publication or display at a public exhibition? Does the Patent Office keep information of the invention secret? Source: Unit IV | SEE Jun 2023 Makeup (18HSI51), Q2a	08	–	–

Publication or public display before filing

An invention must be new on the priority date, that is, not part of the state of the art. Any prior publication, written or oral disclosure, public use or display that makes the knowledge available to the public before the date of filing makes the invention part of the prior art, and it is then **anticipated and cannot be patented**. For example, an inventor was refused a patent on an improved ballpoint pen because he had published its details before filing. Displaying an invention at a public exhibition is a public disclosure of this kind, so the inventor should file the application (at least a provisional specification) **before** any publication or exhibition. Whether any limited statutory exception for notified exhibitions applies is outside the notes and should not be relied upon.

Secrecy in the Patent Office

- An application is not open to the public for a prescribed period after filing; it is then **published**, and the specification and drawings are available to the public on payment of fees. The applicant can request earlier publication.
- The examiner's report is strictly confidential and is not open to inspection even in legal proceedings unless a court directs otherwise.
- If the invention is relevant to **defence**, the Controller may prohibit or restrict publication (a secrecy direction). These are reviewed every six months, no appeal lies, and the application can proceed to acceptance but is not advertised, published or granted while the direction lasts. A defence-relevant invention cannot be filed abroad without the Controller's written permission.

Thus the Patent Office keeps the information confidential only until publication, and in defence cases for as long as the secrecy direction continues.

46	What is the Rice patent? Explain the Indian Government's response to the Rice patent. Source: Unit IV | SEE Apr/May 2024 (21HS51A), Q2b	08	2	2

The Rice (Basmati) patent

RiceTec Inc., a United States firm, was issued US patent number 5663484 on Basmati rice lines and grains on 2 September 1997. The patent covered three things: growing rice plants with characteristics similar to Basmati (semi-dwarf, photoperiod-insensitive, high-yielding, with grains similar or superior to good Basmati), the grain produced by such plants, and a method of selecting rice plants by a starch index (SI) test that predicts the grain's cooking and starch properties.

Response of the Government of India

(1) It announced that it would approach the US Patent Office to re-examine the patent, to protect Indian growers and exporters, and against the use of the Basmati name.

(2) A high-level inter-ministerial group of commerce, industry, external affairs, CSIR, agriculture, biotechnology, APEDA, ICAR and the All India Rice Exporters Association was formed to study the patent in consultation with patent attorneys and agricultural scientists.
(3) The legal strategy, led by the law firm that won the turmeric case, was to challenge the patent on the ground that the plant varieties and grains already exist and therefore cannot be patented.
(4) It built evidence that Basmati-type rice could not be grown in the US: the US rice year book showed that most imported rice was Jasmine rice from Thailand, and the rest came from India and Pakistan, varieties that cannot be grown in the US.
(5) It relied on the earlier turmeric case, in which a US patent on turmeric as a wound-healing agent was revoked after CSIR showed through published evidence that the property had been common knowledge in India for centuries.

Outcome: India won the case. The case shows how traditional knowledge and biodiversity can be exploited through patents elsewhere and the importance of documented prior art.

47	Turmeric is a well-known remedy for various injuries and beauty treatments. State the nature of protection given by law. Source: Unit IV | SEE Mar/Apr 2023 (18HSI51), Q2b	08	–	–

Turmeric is a traditional medicine and beauty remedy known and used in India for centuries. Its protection under law is as follows.

(1) **It cannot be patented by others:** a patent needs novelty and an inventive step. A property known for centuries is part of the prior art and public knowledge, and the mere discovery of a new property or use of a known substance is not an invention. An invention which in effect is traditional knowledge is also not patentable under the Indian Patents Act.
(2) **Defence by prior art:** a United States patent had been granted to two Indian-born scientists on the use of turmeric as a wound-healing agent. CSIR challenged it successfully, producing published evidence that the healing properties of turmeric had been common knowledge in India for centuries. The patent was revoked. US law accepts published or written information from anywhere in the world as prior art, which is why documented evidence won the case.
(3) **Oral knowledge is a weakness:** some national laws, including the US at the time, regarded oral disclosure as prior art only if made in that country, so knowledge handed down orally may be patented abroad.
(4) **Traditional knowledge and biodiversity:** TK of this kind is passed on from generation to generation and cannot easily be protected by conventional patents, whose monopoly is alien to it. It is protected defensively through prior art, post-grant opposition (including knowledge available in a local or indigenous community) and rights such as farmers' and plant breeders' rights.

Conclusion: turmeric is protected not by granting a monopoly to anyone, but by keeping it in the public domain and by documented prior art that defeats any patent claim.

48	Propose a framework for enhancing traditional knowledge protection under international intellectual property laws. Source: Unit IV | CIE-I, 25/11/2024	10	2	4

The problem

- Traditional knowledge (TK) is the knowledge of local communities about plants, animals and life technologies, passed on orally from generation to generation; it evolved under values different from the monopolistic ideas of IP.
- Patent systems judge novelty against prior art and some national laws ignore oral knowledge or knowledge outside their territory, so TK can be patented elsewhere. Industry gets the profits and the communities are treated as suppliers of raw material.
- Access and benefit sharing (ABS) arrangements are themselves premised on IPRs, which are unsuited to biodiversity and TK.

Proposed framework

No.	Element	Measure
1	Recognise TK as prior art	International standard that written, oral and community-held knowledge anywhere in the world counts as prior art against patents.
2	Documentation and databases	Documented, searchable TK records accessible to patent examiners, so applications that claim known TK are refused (as in the turmeric and Basmati cases).
3	Disclosure of origin	Mandatory disclosure of the source and geographical origin of biological material and TK in patent applications; false or missing disclosure is a ground for opposition and revocation, as in the Indian Patents Act.
4	Access and benefit sharing	Prior informed consent of the community and fair sharing of benefits and royalties, not premised on patents alone.
5	Sui generis protection	Special rights for communities, such as farmers' and plant breeders' rights, and effective sui generis systems for plant varieties as TRIPS Article 27 allows.
6	Opposition and revocation	Pre-grant and post-grant opposition in every country on the ground that the claim is anticipated by community knowledge, with legal and financial support to developing countries.
7	Biological material deposit	Deposit of micro-organisms with an international depository under the Budapest Treaty so that the material is traceable.
8	International co-operation	A binding multilateral agreement through the international IP bodies, so that protection does not depend on any one country's law.

Conclusion: the framework combines defensive protection (prior art, opposition), disclosure obligations and fair benefit sharing, so that TK holders are protected and biodiversity-rich countries like India are not exploited.

49	Rahul, an engineer, develops a new water filtration device that purifies water faster than existing technologies. He discovers that a similar product exists in another country but uses a different process. Analyse whether Rahul's invention is patentable under Indian patent law. Discuss the concept, scope and salient features of patents in your answer. Source: Unit IV | Question Bank, Unit 4 Part B Q3	08	3	4

Patentability analysis

- **Novelty:** the invention must not be known anywhere in the world before the filing date. A similar product abroad is prior art: Rahul's product claims succeed only if the device differs from it in a real way. His process is different, so process claims can be novel.
- **Inventive step:** purifying water faster than existing technologies is a technical advance that may not be obvious to a person skilled in the art.
- **Industrial application:** the device can be manufactured and used commercially.
- **Not excluded subject matter:** a device or process is an invention, not a discovery or abstract method.

So Rahul's invention is **patentable in India as a new process (and a new device if it differs from the foreign product)**, if it passes the search for prior art.

Concept, scope and salient features

- **Concept:** a patent gives the inventor the exclusive right to stop others from making, using, selling or importing the invention for 20 years from filing.
- **Scope:** covers products and processes in all fields of technology, and what is protected is what the claims state.
- **Salient features:** novelty, inventive step, industrial application, exclusivity and a limited term of 20 years, and full disclosure in the specification, which Rahul must make.

50	A pharmaceutical researcher wants to patent a herbal remedy for skin diseases that is already widely known in local communities. Is this invention patentable? Explain with reasons and examples of non-patentable inventions. Source: Unit IV | Question Bank, Unit 4 Part B Q4	08	4	4

The herbal remedy is not patentable. It is traditional knowledge already in public use in local communities, so it lacks novelty (it is part of the prior art), and an invention which in effect is traditional knowledge is excluded by the Indian Patents Act. A mere new use or discovery of a known substance is also not an invention.

Examples of non-patentable inventions

- **Natural substances:** discovery of a living or non-living substance occurring in nature (the medicinal property of turmeric).
- **Abstract ideas:** mathematical methods, abstract theories, scientific principles.
- **Contrary to public order or morality, or harmful to life and health.**
- **Artistic creations:** literary, musical and artistic works, covered by copyright.
- **Plants and animals** other than micro-organisms.

Reason: a patent is a monopoly granted for a new technical contribution. Knowledge already in the public domain cannot be monopolised, and such grants would harm local communities.

51	A company designs a software-based algorithm to optimise energy consumption in homes and wants to file a patent. Evaluate whether this invention qualifies for patent protection, considering the patentable criteria. Source: Unit IV | Question Bank, Unit 4 Part B Q5	08	4	4

The question is whether an energy-saving algorithm qualifies for a patent. Under the Indian Patents Act a mathematical or business method, a computer programme per se or an algorithm is not an invention, so **software as such is not patentable**.

Analysis against the criteria

- **Novelty:** the algorithm must not have been published or used anywhere before the filing date.
- **Inventive step:** the method must not be obvious to a person skilled in the field.
- **Industrial application:** the invention must have a practical use such as energy savings in appliances.

If the invention is claimed as a technical method or system, for example a computer programmed to control appliances in a new way to produce an energy saving, it can be patented, because the programmed computer is a modified equipment operating in a new way. If it is only a calculation procedure, it is excluded and the code is protected by copyright instead.

Answer: Patentable only as a technical method or system with a technical effect and industrial application, not as the algorithm per se.

52	A company creates a database documenting all traditional herbal remedies to prevent misuse or unauthorised patents by foreign companies. How does this initiative help in patent law and protection of traditional knowledge? Source: Unit IV | Question Bank, Unit 4 Part B Q7	08	4	4

Documenting traditional herbal remedies in a database is a **traditional knowledge documentation effort**. It helps in patent law and in protecting traditional knowledge in these ways:

(1) **Creates prior art:** a written record makes the knowledge part of the state of the art, so that applications claiming it are anticipated and refused.

(2) **Prevents biopiracy:** foreign companies cannot get patents on knowledge already known, and unauthorised grants can be challenged by pre-grant or post-grant opposition, since knowledge available within a local or indigenous community is a ground.

(3) **Overcomes the oral-knowledge gap:** some national laws do not recognise oral knowledge as prior art; written documentation provides evidence, as the turmeric case showed.

(4) **Protects local communities:** safeguards their rights and keeps ethical use and benefit sharing in view.

(5) **Supports examiners:** it gives patent examiners worldwide a searchable source (as India's Traditional Knowledge

Digital Library does).

Confidential or sacred knowledge can be kept restricted in the database while still being available to patent examiners.

53	An international company develops a method to extract active compounds from a plant traditionally used in Ayurveda, claiming higher efficiency and stability. Can this method be patented, and how does it relate to the protection of traditional knowledge? Source: Unit IV | Question Bank, Unit 4 Part B Q9	08	4	4

Can the method be patented?

Yes, the extraction method can be patented if it is novel (not known from prior art), involves an inventive step (a non-obvious technical improvement giving higher efficiency and stability) and is industrially applicable. A new process that improves the yield or stability of a substance is a typical patentable process invention, as in the drying example where shelf-life improves.

Relation to traditional knowledge

- **The traditional use of the plant cannot be patented:** it is known, an invention which in effect is traditional knowledge is excluded, and a new property or use of a known substance is not an invention.
- **Only the technical improvement is patentable,** such as the specific extraction process; the claim must not cover what the community already knew.
- **Ethical and legal duties:** the application must disclose the source and geographical origin of the biological material, and acknowledgement or benefit sharing with the knowledge holders is expected.
- **Documentation** of the traditional knowledge guards against biopiracy and gives examiners the prior art to test whether the claim is really new.

The law balances the incentive for innovation with the rights of communities and the public interest.

54	Define Traditional Knowledge (TK) and explain its significance in the context of Intellectual Property Rights. Source: Unit IV | Question Bank, Unit 4 Part B Q15	08	3	5

Traditional knowledge (TK) is the knowledge, innovations and practices developed by indigenous and local communities from experience over generations and adapted to local culture and environment, passed on usually by word of mouth. It covers medicinal plants, agriculture and crop adaptation, biodiversity conservation, folklore, handicrafts and other cultural products.

Significance in the IPR context

(1) **Source of valuable leads:** TK guides modern research, especially for pharmaceuticals and agriculture, but is often not recognised as science, and industry takes the rights and profits.

(2) **Prevention of biopiracy:** unauthorised patents on traditional remedies (turmeric, neem, basmati) show the need for protection.

(3) **Conflict with patent concepts:** TK evolved under different values and the monopoly idea of IP is alien to it; patents need novelty judged against prior art, and some laws ignore oral knowledge.

(4) **Cultural heritage and biodiversity:** it keeps community identity and sustainable resource use; India is a mega-diversity country with many crop varieties.

(5) **Economic empowerment and benefit sharing:** access and benefit sharing and sui generis rights such as farmers' and plant breeders' rights give communities a return.

(6) **Legal framework:** in India, the Biological Diversity Act, 2002, the Patents Act (an invention which in effect is traditional knowledge is not patentable) and documentation efforts such as the Traditional Knowledge Digital Library; internationally, the Convention on Biological Diversity and the Nagoya Protocol stress prior informed consent and fair benefit sharing.

55	Discuss the importance of protecting Traditional Knowledge (TK) with reference to relevant case studies on biopiracy. Source: Unit IV | Question Bank, Unit 4 Part B Q16	08	4	5

Meaning and need for protection

Traditional knowledge is the collective know-how of indigenous communities about plants, agriculture and biodiversity. It needs protection:

- to prevent **biopiracy**, the unauthorised use or patenting of TK by corporations without benefit sharing;
- to ensure fair benefit sharing and compensation to the communities;
- to preserve cultural heritage and sustainable use of biodiversity, and to recognise the contribution of indigenous people.

Case studies

(1) **Turmeric case (India versus the US):** a US patent was granted to two Indian-origin scientists for the use of turmeric in wound healing. CSIR challenged it with published evidence that this use had been common knowledge in India for centuries, and the patent was revoked. It set a precedent for rejecting patents on existing TK.
(2) **Neem case (India versus W. R. Grace):** a patent on a method of extracting an antifungal neem oil was revoked by the European Patent Office after India and others showed prior traditional use.
(3) **Basmati case (India versus RiceTec):** RiceTec obtained a US patent on Basmati-like rice lines and grains, the grains and a starch-index selection method. India challenged it on the basis that the varieties already existed, and the claims were not sustained; it highlighted the importance of geographical indication protection.

Protective measures

Documented prior art (the Traditional Knowledge Digital Library), the Biological Diversity Act, 2002, the exclusion of TK from patents, pre-grant and post-grant opposition, and international frameworks such as the Convention on Biological Diversity and the Nagoya Protocol.

Note: The notes give the turmeric and basmati cases; the neem case and the named measures are from the question bank.

56	Explain the procedure for obtaining a patent in India and discuss the rights of a patentee under the Patents Act, 1970. Source: Unit IV | Question Bank, Unit 4 Part B Q17	08	3	5

Procedure for obtaining a patent (summary)

(1) **Patent search (optional but advisable)** to check novelty.
(2) **Filing:** an application with a provisional or complete specification in the prescribed form. A provisional specification fixes the priority date, and the complete specification must follow within 12 months.
(3) **Publication:** the application is published after 18 months from the filing or priority date (earlier on request).
(4) **Examination:** the applicant requests examination (within 48 months of priority), and the examiner checks novelty, inventive step and industrial application. Objections in the examination report are answered within the prescribed time (6 months).
(5) **Opposition:** pre-grant opposition by any person before grant, and post-grant opposition within one year of grant.
(6) **Grant** by the Controller, entry in the Register and publication in the Patent Journal.
(7) **Renewal:** annual renewal fees keep the patent in force for 20 years from filing.

For the full step-wise procedure and the flow chart, see Part B Q40.

Rights of a patentee

(1) **Exclusive right to exploit:** to prevent others from making, using, offering for sale, selling or importing the patented product, or in a process patent from using the process and dealing in the product directly obtained by it, in India, for 20 years.
(2) **Right to assign and licence:** by a written agreement registered with the Controller.
(3) **Right to surrender** the patent by notice to the Controller.
(4) **Right to sue** for infringement in a District Court, and claim an injunction and damages.
(5) **Right of priority:** the Indian filing gives a priority date for filing abroad (convention applications).

Limitations on the rights

- Use by or for the Government (including importing medicines for government hospitals).
- Acquisition of a patent by the Central Government for a public purpose, with compensation.
- Compulsory licences, and secrecy directions for inventions relevant to defence.

57	Explain the concept of compulsory licensing under the Indian Patents Act, 1970 and discuss its importance with the help of a leading case law. Source: Unit IV | Question Bank, Unit 4 Part B Q18	08	4	5

Concept

A **compulsory licence (CL)** is an authorisation granted by the Controller of Patents (or notified by the Central Government) allowing a third party to make, use or sell a patented invention without the patentee's consent. It limits the monopoly, and is based on the principle that patents are granted to encourage invention and to see that inventions are worked in India on a commercial scale and are available to the public at a reasonably affordable price, and not merely to give a monopoly of importation.

Grounds for grant

An application can be made at any time after **three years** from the sealing of the patent, if:

(1) the reasonable requirements of the public with respect to the invention have not been satisfied;
(2) the invention is not available to the public at a reasonably affordable price; or
(3) the invention is not worked in India.

The Controller considers the nature of the invention, the time since sealing, the measures taken by the patentee, the ability of the applicant to work it for public advantage and to take the capital risk, and whether the applicant tried to get a licence on reasonable terms. The Act also provides special compulsory licences on notification by the Central Government in a national emergency, extreme urgency or public non-commercial use (public health crises), and for manufacture and export of patented pharmaceuticals to countries lacking manufacturing capacity. A CL can be terminated when the circumstances end, and a patent can be revoked for non-working after two years from the compulsory licence. The Act's provisions are in Sections 84 to 92.

Importance

- Balances the patentee's rights and the public interest and prevents abuse of the monopoly.
- Ensures access to life-saving medicines and essential technologies at affordable prices.
- Promotes working of the invention in India and domestic manufacturing.

Case: Bayer Corporation v. Natco Pharma

- **Facts:** Bayer held the patent for sorafenib tosylate (Nexavar), a drug for liver and kidney cancer priced at about ₹2,80,000 a month, beyond most patients.
- **Issue:** Natco applied for a compulsory licence under Section 84, arguing that the drug was not available at a reasonably affordable price.
- **Decision:** the Controller granted the licence. Natco had to sell at ₹8,800 a month and pay a 6 per cent royalty to Bayer.
- **Significance:** India's first compulsory licence; it upheld public health and affordability, and showed that patent rights are not absolute.

Note: The Bayer v. Natco case is from the question bank, not the notes.

58	A software company develops an AI-based tool for legal research and seeks to obtain a patent for the invention in India. Examine the patentability of the AI tool under the Indian Patents Act, 1970, with particular reference to the exclusions relating to computer programmes and software. Discuss the major challenges the company may face in obtaining patent protection. Source: Unit IV | CIE-1, 09/10/2026, Q1	10	4	5

A patent is granted only for an **invention**: a new product or process that involves an inventive step and is capable of industrial application, and that does not fall under the list of exclusions in the Act. The AI legal-research tool has to pass both tests.

Patentability of the AI tool

Requirement	Application to the AI tool
Novelty	The tool must not be anticipated by any publication or use anywhere in the world before the filing date. Many legal search engines and text-analysis methods already exist, so the claimed features must be new.
Inventive step	The technical advance must not be obvious to a person skilled in the art. Applying a known machine-learning technique to legal text is likely to be seen as obvious unless it solves a technical problem in a new way.
Industrial application	The invention must be capable of being made or used in a practical activity. A purely intellectual or professional activity (reading and interpreting law) is not industrial in itself, but a working system that processes data in a new way is.
Not an excluded subject	A computer programme per se, an algorithm and a mathematical or business method are not inventions (Section 3(k)); also a mere scheme, rule or mental act (Section 3(m)).

The exclusion relating to computer programmes and software

- An AI tool is essentially software built on algorithms and models. Under Section 3(k) a **computer programme per se, an algorithm or a mathematical or business method** is not patentable. Software as such is protected by copyright (Copyright Act, 1957), not by a patent.
- The exclusion applies to the programme **as such**. A computer system that is programmed to operate in a new way to produce a required result, that is, the computer together with the programme, can be patented because it is a modified machine working in a new way.
- The Indian Patent Office examines such claims for a **technical effect or technical contribution** (an improvement in how the computer or system works, for example in speed, memory use, accuracy of a technical process, or control of hardware), and refuses claims that only automate a business or intellectual procedure.
- A tool whose only result is a better legal answer for the user (finding and ranking case law) gives a result of an intellectual or business nature, which is the weak point of this application.

How the company should draft the claims

- Claim the invention as a **technical system or method**: a processor and memory configured to carry out specified steps (for example a new data structure, indexing or training procedure that reduces processing time) and the technical effect that follows.
- Avoid claims that are only the algorithm, the mathematical model, the rules of legal reasoning or the business method of providing legal advice.
- Protect the source code by **copyright**, and the model weights and training data by **trade secret** and contracts; use the patent for the technical system.

Major challenges the company may face

(1) **Subject-matter exclusion:** objection under Section 3(k) is the most common ground for refusal of AI and software applications.

(2) **Demonstrating technical effect:** the claim must show a technical contribution beyond the working of the programme itself.

(3) **Novelty and inventive step:** crowded prior art in legal-research tools, search engines and natural-language processing makes it hard to show a non-obvious advance.

(4) **Sufficient disclosure:** the specification must describe the invention fully, including the best method; revealing too little of the model, training method or data invites refusal or opposition, while revealing too much gives competitors the secret.

(5) **Inventorship and ownership:** the application must name the true and first inventor, who must be a person; the company needs written assignment from its employee developers, and an output produced by the AI itself cannot be named as inventor.
(6) **Examination, opposition and cost:** the application faces pre-grant and post-grant opposition, delays and examination cost, while AI techniques change faster than the patent is granted.

Answer: The AI tool is not patentable as software or an algorithm per se, but it can be patented if it is claimed as a technical system with a technical effect that is new, inventive and industrially applicable; the code is protected by copyright.

59	A technology company develops an AI-based healthcare application that uses a mathematical algorithm to analyse patient data, presents the results through a graphical interface, and recommends a known medicine for treating a particular disease. The company seeks a patent for the entire system. Analyse the patentability of the invention under the Indian Patents Act, 1970, and identify the relevant categories of non-patentable inventions that may constitute grounds for refusal. Source: Unit IV | CIE-1, 09/10/2026, Q3	10	3	5

The company claims the **entire system**, which has three elements: a mathematical algorithm that analyses patient data, a graphical interface that presents the results, and a recommendation of a known medicine for a disease. A patent can be granted only for an invention that is new, involves an inventive step, is capable of industrial application and is not excluded by Section 3 of the Patents Act, 1970. Each element has to be examined separately.

Analysis of the invention

Element	Ground for refusal	Reason
Mathematical algorithm that analyses patient data	Mathematical method, algorithm or computer programme per se (Section 3(k))	The algorithm and the programme that executes it are not inventions. They can be allowed only as part of a system that produces a technical effect.
Graphical interface showing the results	Mere presentation of information (Section 3(n))	How the results are displayed adds no technical solution; it is not a technical solution.
Recommendation of a known medicine for a disease	Mere new use of a known substance (Section 3(d)); method of medical treatment or diagnosis (Section 3(i))	The medicine is already known and is not changed, so there is no new substance or enhanced efficacy. A method of prescribing or treating human beings is excluded as a matter of health policy.
Underlying analysis of patient data	Mere scheme, rule or method of performing a mental act (Section 3(m))	If the system only mimics the reasoning of a doctor, it is a mental act performed by machine.

Conclusion on the system as a whole

- The patentability of a combination is not decided by adding up its parts: each excluded feature contributes nothing to the novelty or inventive step, and a claim to the system as a whole is refused if its only inventive content lies in the excluded subject matter.
- The system is patentable only if the company can show a **technical contribution** in the way the data are processed, for example a new method of acquiring or processing signals from a medical device that gives a technical result such as faster or more accurate processing, and claims it as a computer system operating in a new way.
- It must also be shown that the technical solution is **new** (not disclosed in prior art), **non-obvious** and **industrially applicable**, with a full description in the specification.
- Claims reciting the recommendation of a known medicine, the display of results or the algorithm per se should be removed; the software is protected by copyright.

Relevant categories of non-patentable inventions that may be grounds for refusal

(1) Mathematical method, computer programme per se or algorithm.
(2) Mere presentation of information.
(3) Mere new use of a known substance, or mere new form of a known substance without enhanced efficacy.
(4) Medical treatment, diagnosis or therapeutic methods for human beings.

(5) Mere scheme, rule or method of performing a mental act.
(6) Also, any invention contrary to public order or morality or seriously harmful to human health, if the application recommends a harmful treatment.

Answer: The system as claimed is likely to be refused; only a claim to a technical system with a new, non-obvious technical effect, without the algorithm, interface and known-medicine elements, can be patented.

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